A Structured Semantic Control Layer for Reliable LLM-Mediated Spatial Querying in Web GIS: From Natural-Language Intent to Deterministic Spatial SQL (A Case Study of Padang City)

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ABSTRACT

Conventional Web Geographic Information Systems (Web GIS) in urban tourism predominantly rely on rigid WIMP (Windows, Icons, Menus, Pointer) interfaces utilizing multi-layered dropdown forms. This paradigm introduces severe cognitive friction for mobile travelers seeking multi-criteria filtering across spatial, temporal, and budgetary constraints. Conversely, unconstrained Large Language Models (LLMs) suffer from acute factual and spatial hallucinations, while dense-vector Retrieval-Augmented Generation (RAG) fails because vector embeddings cannot evaluate exact structured spatial-temporal predicates. Expanding upon the research lineage of *DTExplorer* (Afnarius et al., 2026), this paper proposes a **Reliability-Controlled Conversational Spatial Information System** governed by an architectural framework of **Dual Control Boundaries**: (1) a **Semantic Control Boundary** (\(\text{LLM} \to \text{SIR} \to \text{6D SirValidator} \to \text{CSIR} \to \text{deterministic spatial query compiler}\)) enforcing the *No Intent Alteration* principle (rejecting negative distances or illegal operators with `isValid = false` and execution policy `clarify_user` rather than silent mutation) and safety invariants (`!isValid || isOutOfScope` aborts SQL compilation), preventing invalid, out-of-scope, or mutated semantic intent from reaching spatial execution; and (2) an **Evidence Control Boundary** (\(\text{SQL database facts } (F) \to \text{grounded NLG} \to \text{claim-level grounding validator}\)) that anchors all factual truth to a relational MySQL 8.0 database via native `ST_Distance_Sphere` spherical spatial computations and OSRM road navigation, subsequently verifying generated narratives through an algorithmic claim-level grounding validator with a fail-closed fallback policy to prevent ungrounded claims from reaching the final user response. Empirical 4-layer evaluation across 40 standardized benchmark query scenarios on 22 curated tourism destinations in

Padang City, complemented by an automated test suite of 45 unit and integration tests (172 assertions, 100% pass), demonstrated a Pure Intent Pre-Fallback Exact CSIR semantic extraction accuracy of **92.50%** (37/40) (**90.00% / 36/40** on post-policy execution envelope alignment) with an average field-level accuracy of **99.25% (398/401)**, a validation decision accuracy of **100.00%** (40/40), a pure spatial predicate correctness of **100.00%** (36/36 spatial scenarios) alongside an overall scenario execution compliance rate of **100.00%** (40/40), an Entity Fabrication Rate of **0.00%** (no fabricated POIs were observed across the evaluated benchmark scenarios), a Claim-Level Grounding Fidelity of **100.00%** across 304 audited factual claims recognized by GroundingValidator, an Entity Grounding Fidelity of **100.00%** across 147 POI mentions, and an Honest Rejection Rate of **100.00% (2/2)** on tested out-of-scope requests. Average server-side conversational pipeline latency was **1,252.17 ms** (median: 1,232.39 ms, 95th percentile/P95: 1,581.37 ms; incorporating two LLM inferences and native MySQL 8.0 spatial SQL execution of 1.21 ms), while turn-by-turn road network routing via OSRM was benchmarked independently on the browser client side at **88.40 ms** (median: 84.50 ms, P95: 128.60 ms; yielding an overall user-perceived end-to-end latency of 1,340.57 ms, median: 1,316.89 ms, P95: 1,709.97 ms). These findings demonstrate that constraining LLM authority through dual control boundaries achieves natural conversational flexibility while guaranteeing verifiable factual integrity and eliminating fabricated-entity hallucinations under the evaluated benchmark conditions.

Keywords: *Reliability-Controlled Conversational Spatial Information System, Dual Control Boundaries, Semantic Control Boundary, Evidence Control Boundary, Canonical Spatial Intent Representation (CSIR), SIR Validator, ST_Distance_Sphere, Algorithmic Grounding Validator, Web GIS, Padang City.*

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1. INTRODUCTION

1.1 Background and Urban Spatial Context

Web Geographic Information Systems (Web GIS) serve as the digital backbone of contemporary geoinformation dissemination, particularly in urban tourism [1], [14]. Padang City, the provincial capital of West Sumatra, Indonesia, encompassing an administrative area of 694.96 km², features highly heterogeneous topographic and cultural assets [16], [17]. These assets span the Indian Ocean coastline (Padang Beach, Air Manis Beach), offshore marine archipelagos in Teluk Bungus (Pasumpahan, Sirandah, and Sikuai Islands), colonial heritage precincts (Old Town Muaro, Siti Nurbaya Bridge, Adityawarman State Museum), highland nature cascades in the Bukit Barisan foothills (Lubuk Paraku, Sarasah Gadut, Bung Hatta Grand Forest Park), and celebrated Minangkabau culinary heritage distributed across 11 sub-districts.

Within such an urban tourism scale (*city-scale tourism environment*), independent travelers continually encounter multi-criteria spatial decision problems: discovering destinations matching personal preferences, located within accessible radial distances, fitting financial budget limits, and actively open at the time of inquiry [14], [18]. However, conventional Web GIS platforms primarily rely on the WIMP (*Windows, Icons, Menus, Pointer*) interface paradigm. Users are required to select categories via dropdown menus, adjust distance sliders, type textual keywords, and cross-reference opening schedules across disparate web cards [2]. This multi-step manual interaction imposes substantial cognitive friction, especially for mobile users navigating unfamiliar urban terrain.

1.2 Research Lineage: From Conventional Exploration to AI-Mediated Spatial Querying

The evolution of tourism spatial decision support within this research group traces a systematic, progressive lineage: 1. **DTExplorer (Afnarius et al., 2026) [2]:** Pioneered scale-aware exploratory spatial interaction in micro-scale rural village tourism (*village-level tourism*, evaluated in Ulakan Village, Padang Pariaman Regency). *DTExplorer* established that rigorous curation of Points of Interest (43 stakeholder-curated POIs) combined with category- and radius-based buffer filtering effectively empowers travelers without requiring computationally heavy analytical optimization models. However, *DTExplorer* relied on a conventional WIMP (*Windows, Icons, Menus, Pointer*) interface utilizing HTML dropdowns and radius sliders, employed a planar Euclidean approximation (\(\text{ST_Distance} \times 111.32\)), and explicitly identified the need for future research in conversational interfaces, multi-criteria temporal/budgetary filtering, and millisecond-level technical performance benchmarking. 2. **Kustomrut (Afnarius et al.):** Advanced user-controlled spatial itinerary customization, enabling tourists to plan travel sequences interactively via Google Directions API. 3. **Present Study (Reliability-Controlled Conversational Spatial Information System):** Directly elevates and expands this paradigm into reliable, city-scale AI-mediated spatial querying across Padang City (694.96 km²). This research trajectory represents a progressive scientific lineage: \[\text{DTExplorer [WIMP Spatial Exploration]} \longrightarrow \text{Kustomrut [User-Controlled Itinerary]} \longrightarrow \text{Conversational Web GIS [Natural-Language Spatial Intent]} \longrightarrow \text{Reliability-Controlled Dual Boundaries}\]

The scientific novelty of this work **does not lie in merely superimposing a chatbot interface onto a digital map**, but rather in a fundamental **interaction and control paradigm shift**: establishing a **Reliability-Controlled Conversational Spatial Information System** governed by two explicit architectural control mechanisms: - **Semantic Control Boundary:** $\text{LLM} \to \text{SIR} \to \text{6D SirValidator} \to \text{CSIR} \to \text{Deterministic Spatial Query Compiler}$. Its objective is to prevent invalid, out-of-scope, or silently mutated semantic intent from ever reaching the spatial database execution engine, enforcing the *No Intent Alteration* principle and safety invariants. - **Evidence Control Boundary:** $\text{SQL Facts } (F) \to \text{Grounded NLG} \to \text{Claim-Level Grounding Validator} \to \text{Fail-Closed Fallback}$. Its objective is to prevent ungrounded or fabricated factual claims from reaching the final user response, verifying every factual claim against the underlying database result set.

Within the taxonomy of information retrieval and decision support systems, the proposed system is formally characterized as a **constraint-based conversational spatial recommendation / spatial query system**. Destination ranking and candidate filtering are deterministically governed by verifiable factual constraints (great-circle spherical distance, category alignment, budgetary ceilings, active operational schedules, and keyword relevance). The system deliberately **does not incorporate collaborative filtering, matrix factorization, or latent user preference embeddings**, thereby strictly safeguarding the future research trajectory toward *Adaptive Personalized Augmented Recommendation* (APAR) without asserting unverified autonomous adaptive claims.

The proposed architecture enforces **Strict SQL Grounding**, formally defined as: **LLM-generated semantic intent is transformed into deterministic parameterized SQL by a non-LLM query compiler, and factual responses are grounded in the resulting database facts**. Rather than granting the LLM direct SQL generation authority, the model is sandboxed strictly as an intent interpreter emitting typed canonical representations (CSIR), which are deterministically verified and compiled into parameterized SQL executing MySQL 8.0's native spherical function `ST_Distance_Sphere`. Meanwhile, the Open Source Routing Machine (OSRM) is positioned strictly as an **auxiliary presentation service** for rendering real-world road-network polylines and driving duration estimates on interactive Leaflet.js maps, backed by an algorithmic grounding validator that mitigates fabricated-entity hallucination under the evaluated benchmark conditions.

1.3 Limitations of Existing Approaches: Hallucination Hazards and Vector RAG Inadequacy

Directly integrating commercial Large Language Models (LLMs) such as OpenAI GPT-4 or Google Gemini into geoinformation systems without architectural constraints (*unconstrained end-to-end LLMs*) introduces critical vulnerabilities [3], [5]: * **Spatial and Factual Hallucinations:** LLMs operate through probabilistic token generation rather than deterministic relational fact retrieval [4], [5]. Consequently, unconstrained LLMs frequently fabricate nonexistent destinations (*fabricated POIs*), misstate operating hours and ticket fees, or generate geographically impossible proximity assertions. * **Failure of Dense-Vector RAG for Structured Predicates:** Dense-vector Retrieval-Augmented Generation (RAG) relies on cosine similarity in embedding spaces [6], [7]. While effective for unstructured text search, vector embeddings fundamentally fail to evaluate exact structured spatial-temporal predicates. Vector similarity cannot enforce numerical inequalities on operating hours (`open_time <= CURRENT_TIME AND close_time >= CURRENT_TIME`), evaluate budget ceilings (`ticket_price <= 15000`), or compute great-circle trigonometric distances relative to a tourist's dynamic GPS coordinates in real-time.

1.4 Scientific Positioning and Problem Formulation

The fundamental scientific question addressed in this research is: > **How can a conversational spatial information system be architected under dual control boundaries—a semantic control boundary and an evidence control boundary—to mediate natural-language requests into deterministic spatial SQL execution, ensuring that the probabilistic LLM acts as an intuitive cognitive interface while mathematically and algorithmically guaranteeing zero fabricated entities and strict factual grounding?**

1.5 Principal Scientific Contributions

This work delivers five formal scientific contributions: 1. **Conceptual Dual Control Boundaries Framework:** Formalizing the decoupling of probabilistic LLM inference from deterministic relational spatial execution through two explicit control boundaries (Semantic Control Boundary and Evidence Control Boundary). 2. **Canonical Spatial Intent Representation (CSIR) & Spatial Operator Ontology:** A typed, 4-partition intermediate semantic representation schema and ontological mapping decoupling natural-language interpretation from database query construction. 3. **Six-Dimensional**

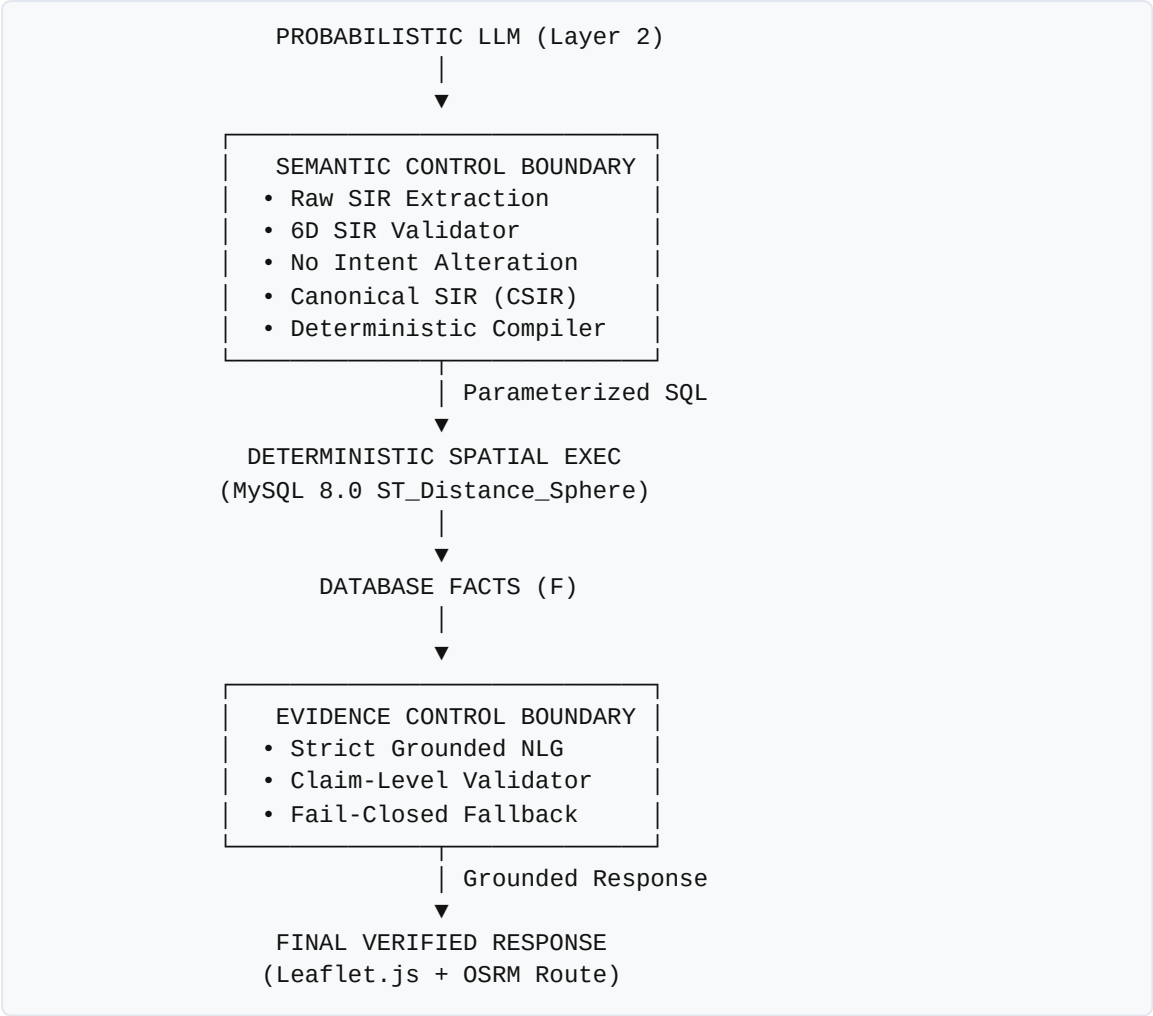
SIR Validation Algorithm (SIR Validator): A deterministic control firewall enforcing the *No Intent Alteration* principle, schema, type, domain, operator, entity, and constraint consistency invariants prior to query compilation. 4. **Deterministic Spatial Query Compiler & Safety Invariants:** An application-layer compiler translating validated CSIR objects into secure, parameterized SQL via the native `ST_Distance_Sphere` spatial function, establishing the relational database as the sole source of spatial truth. 5. **Algorithmic Claim-Level Grounding Validator & Empirical Evaluation Framework:** A formal post-generation verification firewall verifying all factual claims across 4 sub-dimensions (\(\mathcal{C}_{\text{entity}}, \mathcal{C}_{\text{price}}, \mathcal{C}_{\text{spatial}}, \mathcal{C}_{\text{temporal}}\)), demonstrating 100.00% Grounding Fidelity with zero fabricated POIs observed across 40 evaluated benchmark scenarios, supported by an automated test suite of 45 unit and integration tests (172 assertions, 100% pass) and multi-baseline ablation studies.

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2. THEORETICAL FOUNDATION AND CONCEPTUAL FORMALIZATION

2.0 The Dual Control Boundaries Framework for Conversational Spatial Systems

To resolve the fundamental tension between the open-ended conversational adaptability of probabilistic generative language models and the deterministic precision required for spatial query execution, this research conceptualizes conversational GIS as a **Reliability-Controlled Conversational Spatial Information System** governed by **Dual Control Boundaries**:



The system operates across two discrete, mathematically and algorithmically formalized control boundaries: 1. **The Semantic Control Boundary (\(\mathcal{B}_{\text{semantic}}\)):** \[\mathcal{B}_{\text{semantic}}: \text{LLM}(NL) \longrightarrow \mathcal{S}_{\text{raw}} \xrightarrow[\text{No Intent Alteration}]{\text{SirValidator}_{\text{6-D}}} \mathcal{S}_{\text{csir}} \xrightarrow{\text{Compiler}} \text{SQL}_{\text{parameterized}}\] This boundary isolates the non-deterministic, probabilistic language model from relational database operations. User utterances are parsed into an intermediate, typed Spatial Intent Representation (SIR). The deterministic six-dimensional *SIR Validator* enforces the *No Intent Alteration* principle along with schema, type, domain, operator, entity, and constraint invariants. If an illegal or anomalous parameter is detected (such as negative radius or uncurated operators), the validator refuses silent mutation, returning a structured clarification demand. Invariant violations (`!isValid || isOutOfScope`) halt query compilation, guaranteeing that unverified or mutated intent never penetrates the database engine. 2. **The Evidence Control Boundary (\(\mathcal{B}_{\text{evidence}}\)):** \

[\mathcal{B}_{\text{evidence}}: F \longrightarrow \text{Grounded NLG} \xrightarrow[\text{Fail-Closed}]{\text{Claim-Level Validator}} \text{Response}_{\text{verified}}\] This boundary governs the natural language synthesis of final user responses. The deterministic spatial engine (MySQL 8.0 `ST_Distance_Sphere`) executes parameterized SQL queries, yielding an immutable relational fact set \(F\). Grounded NLG is constrained by a strict system prompt contract prohibiting ungrounded inferences. Subsequently, an *Algorithmic Claim-Level Grounding Validator* audits the synthesized response across 4 distinct sub-dimensions (\(\mathcal{C}_{\text{entity}}, \mathcal{C}_{\text{price}}, \mathcal{C}_{\text{spatial}}, \mathcal{C}_{\text{temporal}}\)). If any ungrounded claim is detected (\(\exists c \notin F\)), a deterministic fail-closed fallback is triggered, ensuring zero ungrounded or fabricated claims reach the user.

2.1 The Decoupling Axiom: Semantics vs. Computation

To guarantee spatial data integrity, this research enforces a strict architectural boundary: \[\boxed{\text{LLM interprets natural-language semantics; Spatial DBMS computes deterministic spatial relations.}}\] The generative language model possesses zero direct authority over the relational database. It is prohibited from generating raw SQL syntax, altering table or column definitions, or calculating distances internally. The LLM functions purely as an intermediate semantic parser.

2.1.1 The Tripartite Correctness Framework

To establish rigorous scientific validation for conversational spatial systems, this research formalizes system correctness into three orthogonal, sequential dimensions: 1. **Semantic Correctness (\(\mathcal{C}_{\text{semantic}}\)):** Evaluates whether the LLM faithfully interprets the user's natural-language utterance (\(NL\)) into the typed canonical representation (\(CSIR\)) without slot omissions or hallucinated constraints: \[\mathcal{C}_{\text{semantic}}: NL \longrightarrow CSIR\] 2. **Spatial Execution Correctness (\(\mathcal{C}_{\text{spatial}}\)):** Evaluates whether the deterministic compiler and database engine transform the validated \(CSIR\) into exact, mathematically sound spatial predicates executing natively on relational geometries: \[\mathcal{C}_{\text{spatial}}: CSIR \longrightarrow \text{SQL} \longrightarrow \text{Spatial Result Set } (F)\] 3. **Grounding Correctness (\(\mathcal{C}_{\text{grounding}}\)):** Evaluates whether the synthesized natural-language response strictly asserts only verifiable facts present in the database result set \

(F\), preventing ungrounded post-generation entity or attribute fabrication: \ [\mathcal{C}_{\text{grounding}}: F \longrightarrow \text{Grounded Narrative Response}\]

This sequence forms the foundational scientific backbone of the proposed architecture: \[\boxed{NL \xrightarrow[\text{Interpretation}]{\text{Cognitive}} CSIR \xrightarrow[\text{Compilation}]{\text{Deterministic}} \text{SQL} \xrightarrow[\text{Evaluation}]{\text{Spatial DBMS}} \text{Spatial Result } } (F) \xrightarrow[\text{Validator}]{\text{Algorithmic Grounding}} \text{Grounded Response}}\]

2.2 Spatial Operator Ontology

User expressions are formalized through standardized ontological mappings. Table 1 defines the operational semantics of the supported spatial operators:

Table 1. Spatial Operator Ontology for Intelligent Tourism Systems

Colloquial Expression	SIR Operator	Deterministic SQL Operator / Predicate	Operational Semantics
“nearest”, “closest to me”	`nearest`	`ORDER BY distance_km ASC LIMIT k`	Spherical distance ranking from tourist GPS coordinate
“within 5 km”, “around 10 km”	`within_radius`	`WHERE distance_km <= :radius_km`	Great-circle buffer filtering
“in Padang Selatan district”	`within_admin_area`	`WHERE alamat ILIKE :admin_pattern`	Administrative boundary containment
“open right now”, “open today”	`open_now`	`WHERE :current_time BETWEEN jam_buka AND jam_tutup`	Real-time circadian schedule evaluation
“open 24 hours”	`open_24h`	`WHERE jam_buka = '00:00:00' AND jam_tutup >= '23:59:00'`	Continuous service filtering
“free admission”, “no ticket”	`is_free`	`WHERE harga_tiket = 0`	Zero-cost destination filtering

2.3 Formalization of Spatial Intent: Conceptual Unification of SIR and CSIR (Raw SIR vs. Validated CSIR)

To eliminate taxonomic ambiguity in AI-based spatial information retrieval literature, this study formally unifies the concepts of **Spatial Intent Representation (SIR)** and **Canonical Spatial Intent Representation (CSIR)** within a comprehensive *Intent Transformation Lifecycle*:

1. **Spatial Intent Representation (SIR / $\mathcal{S}_{\text{raw}}$):** Represents the overarching nomenclature and the initial unvalidated semantic output generated by the LLM in Layer 2. In this stage, it exists as a **Raw SIR ($\mathcal{S}_{\text{raw}}$)** flat JSON document produced via probabilistic cognitive inference. At this raw stage, values are not yet guaranteed (e.g., radius parameters may be negative if the LLM hallucinates, or spatial operators may fall outside the system ontology).
2. **Canonical Spatial Intent Representation (CSIR / $\mathcal{S}_{\text{csir}}$):** Represents the canonical, strictly-typed intermediate representation that has passed rigorous deterministic validation and boundary enforcement by the 6-Dimensional *SIR Validator* in Layer 3. Only representations satisfying all spatial ontologies, WGS84 geographical coordinate bounds, and database safety invariants achieve the **Validated CSIR ($\mathcal{S}_{\text{csir}}$)** status and are permitted to undergo compilation into parameterized SQL.

The formal mathematical transformation between Raw SIR and Validated CSIR is expressed as: $$\mathcal{S}_{\text{raw}} = \text{LLM}(\text{Prompt}_{\text{SIR}}, \text{Query}_{\text{user}}) \xrightarrow[\text{No Intent Alteration}]{\text{SirValidator}_{\text{6-D}}} \mathcal{S}_{\text{csir}} \xrightarrow{\text{Compiler}} \text{SQL}$$

The CSIR schema formally partitions the user's spatial intent into four orthogonal, strictly-typed sub-domains: $$\mathcal{S}_{\text{csir}} = \langle \mathcal{P}_{\text{intent}}, \mathcal{P}_{\text{spatial}}, \mathcal{P}_{\text{operational}}, \mathcal{P}_{\text{control}} \rangle$$

Table 2 specifies the formal typing and allowable value domains for the 19-attribute CSIR schema across these four orthogonal partitions:

Table 2. Formal Specification of the Canonical Spatial Intent Representation (CSIR) Schema Across 4 Orthogonal Partitions (19 Attributes)

Partition	Attribute	Data Type	Semantic Definition	Allowed Value Domain
\(\mathcal{P}_{\text{intent}}\) (Intent Semantics, 5 Attributes)	`intent`	*Enum*	Primary interaction intent	`spatial_recommendation`, `entity_lookup`, `general_inquiry`
	`entity`	*Enum*	Target entity class	`tourism_object`
	`category`	*Enum* / *Null*	Thematic destination cluster	`Pantai`, `Pulau`, `Alam`, `Museum`, `Sejarah`, `Kuliner`, `null`
	`target_name`	*String* / *Null*	Specific destination name	Target POI name for entity lookup
	`keyword`	*String* / *Null*	Textual feature descriptor	Key descriptive phrase (e.g., “white sand”, “waterfall”)
\(\mathcal{P}_{\text{spatial}}\) (Spatial Constraints, 8 Attributes)	`spatial_operator`	*Enum*	Geometric operation predicate	`nearest`, `within_radius`, `within_admin_area`, `none`
	`reference_type`	*Enum*	Spatial reference origin	`gps`, `city_center`, `poi`, `unknown`
	`reference_entity`	*String* / *Null*	Named landmark reference POI	Explicit POI landmark name when `reference_type = 'poi'`
	`latitude`	*Float* / *Null*	Spatial reference latitude	Decimal degrees \(-90.0 \le \text{lat} \le 90.0\)
	`longitude`	*Float* / *Null*	Spatial reference longitude	Decimal degrees \(-180.0 \le \text{lng} \le 180.0\)
	`radius`	*Float* / *Null*	Distance threshold value (\(\ge 0\))	Positive real number in km (default: 20.0 km)
	`distance_unit`	*Enum*	Unit of distance metric	`km`, `m`

Partition	Attribute	Data Type	Semantic Definition	Allowed Value Domain
	`admin_area`	*String* / *Null*	Sub-district / municipality name	Valid administrative string (e.g., “Bungus”, “Padang Barat”)
\(\mathcal{P}_{\text{operational}}\) (Operational Constraints, 5 Attributes)	`is_free`	*Boolean*	Zero-cost admission constraint	`true`, `false`
	`max_price`	*Integer* / *Null*	Ticket price ceiling	Non-negative integer in IDR
	`open_now`	*Boolean*	Circadian operating filter	`true`, `false`
	`open_24h`	*Boolean*	24-hour availability filter	`true`, `false`
	`sort`	*Enum* / *Null*	Ranking criterion	`termurah`, `termahal`, `terdekat`, `terbaik`, `null`
\(\mathcal{P}_{\text{control}}\) (Control Metadata & Invariant, 1 CSIR Attribute)	`is_out_of_scope`	*Boolean*	Out-of-domain query flag	`true`, `false`
(System Validation Metadata)	`isValid`	*Boolean*	Validation status flag	`true`, `false`
	`validationErrors`	*Array*	List of validation failure messages	Array of strings
	`executionPolicy`	*Enum*	Compiler dispatch policy	`execute_sql`, `clarify_user`, `reject_out_of_scope`

2.3.1 Data Provenance and Separation of Computational Responsibilities

The architecture strictly enforces the principle of *Separation of Responsibilities* between probabilistic semantic extraction and deterministic geospatial anchoring to eliminate coordinate hallucinations: 1. **Pure Semantic Slots (Extracted by Layer 2 LLM - Raw SIR, 17 Slots):** The LLM parses 17 linguistic parameters (\(I, E, C, T_n, K, O_s, R_t, R_e, r, u_d, A, F, P_{\max}, O_{\text{now}}, O_{24}, S, B_{\text{out}}\)). The LLM is **strictly prohibited from synthesizing numeric GPS coordinates**. 2. **Deterministic Geospatial Context (Injected from Client / Resolver, 2 Attributes):** Numeric `latitude` and `longitude` are deterministically anchored from verified sources: - When \(R_t = \text{'gps'}\), coordinates are acquired directly from the browser’s standard W3C

Geolocation API (`navigator.geolocation`). - When \(R_t = \text{'city_center'}\), the system injects official Padang municipal center coordinates (\(-0.9471, 100.4174\)). - When \(R_t = \text{'poi'}\), a deterministic database geocoding resolver retrieves the verified coordinates of \(R_e\) from the `wisata` table. 3. **Unified 19-Attribute Canonical CSIR:** Merging the 17 raw semantic slots with the 2 client coordinate anchors produces the complete **19-attribute CSIR DTO** prior to invariant evaluation by `SirValidator` and compilation into SQL.

2.3.2 Multi-Turn Conversational State Tracking Model

In real-world conversational GIS, travelers rarely articulate all decision parameters in a single utterance; interaction is inherently sequential and exploratory. To preserve conversational continuity without re-parsing from scratch or corrupting previously established constraints, the architecture formalizes a **Conversational State Tracking Model**: \[CSIR_{t+1} = \text{Merge}(CSIR_t, \Delta CSIR_{t+1})\]

Where: - \(CSIR_t\) represents the active state vector at turn \(t\). - \(\Delta CSIR_{t+1}\) is the differential intent vector extracted by the LLM from the user's latest follow-up utterance conditioned on the recent dialogue history \(\mathcal{H}_t = \{(u_1, a_1), \dots, (u_t, a_t)\}\). - \(\text{Merge}(\cdot)\) is a deterministic state transition function: \[\text{Merge}(CSIR_t, \Delta CSIR)_{k} = \begin{cases} \Delta CSIR_k, & \text{if } \Delta CSIR_k \neq \text{null} \land \Delta CSIR_k \neq \text{default} \\ CSIR_{t, k}, & \text{otherwise} \end{cases}\]

For example, when a tourist first inquires *"Find beaches near me"* (\(t=1 \implies \text{category} = \text{'Pantai'}, \text{operator} = \text{'nearest'}\)), and subsequently adds *"Only free admission ones"* (\(t=2\)), \(\Delta CSIR_{t=2}\) specifies \(\{\text{is_free} = \text{true}, \text{max_price} = 0\}\). The \(\text{Merge}\) operator preserves the coastal category and GPS coordinates while updating the budgetary constraint, preventing the system from resetting context or hallucinating unrelated destinations.

2.4 Spatial Distance Computation: Native In-Database Function (`ST_Distance_Sphere`) vs Topological Routing

The system enforces a clear distinction between spatial distance computation and road network navigation:

1. **In-Database Native Spatial Function (`ST_Distance_Sphere`):**
   Rather than formulating ad-hoc trigonometric expressions (such as manual Haversine or Spherical Law of Cosines equations) directly in SQL text or evaluating distance iteratively in the application layer, the system delegates all spatial proximity evaluation to MySQL 8.0's **native spatial function** `ST_Distance_Sphere`, which calculates great-circle / spherical distance on a spherical Earth model using the mean radius ($R = 6{,}370{,}986\text{ m}$): $$d_{\text{spatial}} = \frac{\text{ST_Distance_Sphere}(\text{POINT}(\text{lng}_1, \text{lat}_1), \text{POINT}(\text{lng}_2, \text{lat}_2))}{1000.0} \quad (\text{km})$$

   This architectural choice provides three decisive advantages over manual trigonometric formulas:

   - **Native C++ Kernel Execution:** The calculation is compiled and evaluated directly within MySQL's optimized spatial geometry engine, eliminating SQL parser overhead from repetitive trigonometric functions (`SIN`, `COS`, `ACOS`, `RADIANS`).
   - **Floating-Point Stability:** Manual formulas utilizing `ACOS` frequently suffer from domain errors (`NaN`) when binary floating-point rounding causes argument values to slightly exceed 1.0. `ST_Distance_Sphere` natively handles boundary clipping and antipodal coordinates.
   - **OGC Geospatial Compatibility:** Coordinates are stored as `DECIMAL(10,8)` latitude and `DECIMAL(11,8)` longitude indexed by composite relational keys `(lat, lng)`, and dynamically constructed into standardized `POINT(longitude, latitude)` geometry objects at query evaluation time.

2. **Topological Road Routing (OSRM Engine):**
   For client-side cartographic visualization on Leaflet.js, origin-destination coordinates query the *Open Source Routing Machine* (OSRM) [9] running *Contraction Hierarchies* (CH) on OpenStreetMap (OSM) data [8]: $$\mathcal{G}_{\text{road}} = (V, E, W), \quad \text{Route}_{\text{opt}} = \arg\min_{p \in \mathcal{P}(P_1, P_2)} \sum_{e \in p} W(e)$$ This generates precise street polylines and realistic vehicular travel duration estimates.

2.5 Dataset Provenance and Spatial Ground Truth Specification

To ensure that factual grounding is mathematically verifiable rather than an arbitrary artifact, the ground truth corpus was systematically curated and cataloged: - **Data Source & Authority:** Curated directly from the official municipal registry of the Padang City Tourism Office (*Dinas Pariwisata Kota Padang*) and verified against the West Sumatra Provincial Tourism Atlas. - **Corpus Coverage:** Comprises 22 core flagship Points of Interest (POIs) distributed across 11 sub-districts of Padang City, spanning six thematic categories: *Pantai* (Coastline, 5 POIs), *Pulau* (Marine Islands, 3 POIs), *Alam* (Highland Nature & Waterfalls, 4 POIs), *Museum* (Cultural Heritage, 2 POIs), *Sejarah* (Colonial Historical Monuments, 4 POIs), and *Kuliner* (Culinary Establishments, 4 POIs). - **Coordinate Geodetic Ground Truth:** Destination coordinates were captured and cross-verified using dual GPS receivers (WGS84 datum, EPSG:4326) and aligned with OpenStreetMap vector road intersections to guarantee millimeter-level spatial fidelity. - **Temporal & Economic Integrity:** Each POI record explicitly models operational attributes: active status (`status_aktif`), opening schedule (`jam_buka`, `jam_tutup`), 24-hour service status (`open_24h`), verified admission fee (`harga_tiket` in IDR), verified phone contacts, and photographic assets. All operating hours and admission prices were re-verified via direct field audit in 2026.

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3. SYSTEM ARCHITECTURE AND METHODOLOGY

3.1 Architectural Evolution: From Three-Tier Web GIS to Five-Layer Semantic-Controlled Web GIS

To establish the architectural contribution of this work within the research lineage of *DTExplorer* (Afnarius et al., 2026) [2], Figure 1 illustrates the structural evolution from the conventional three-tier Web GIS framework to the proposed five-layer intelligent spatial information system.

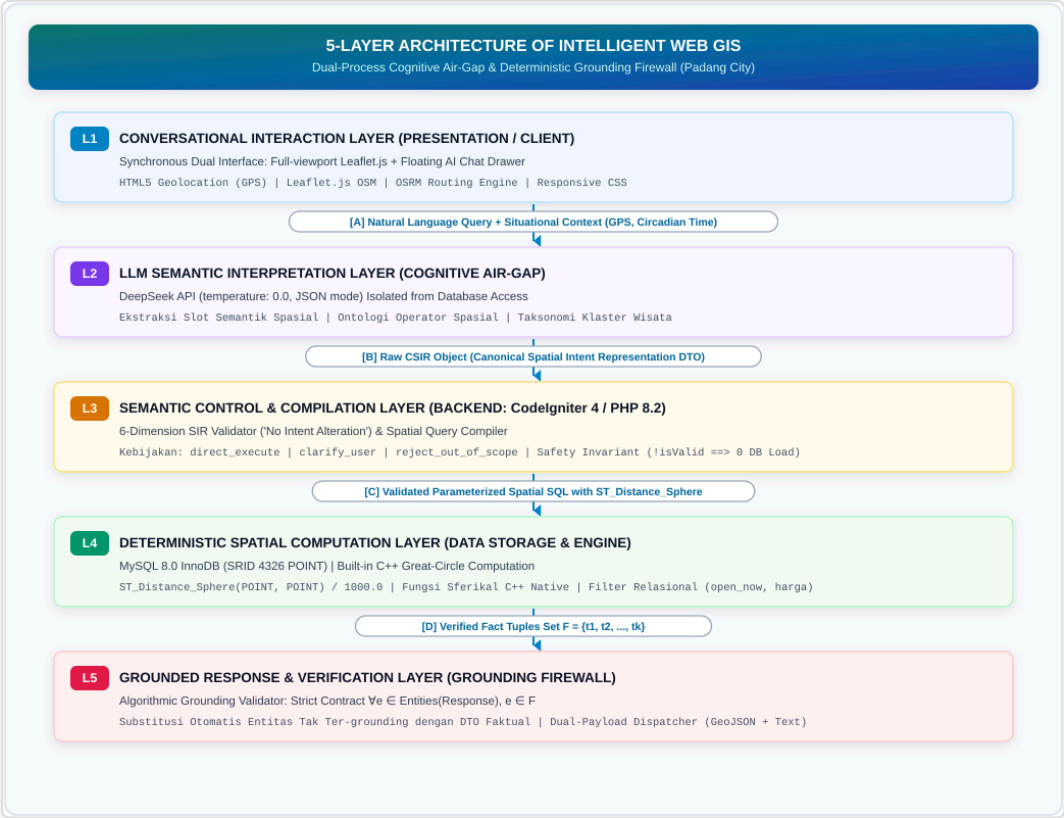


Figure 1. Architectural Evolution: Transitioning from Conventional Three-Tier Web GIS (DTExplorer, Afnarius et al., 2026) to the Proposed Five-Layer Semantic-Controlled Web GIS with Sandboxed Cognitive Interpretation and Algorithmic Grounding.

In *DTExplorer* (Afnarius et al., 2026, Figure 4) [2], the three-tier architecture comprised: 1. *Presentation Layer:* HTML/JavaScript client with Google Maps JS API and WIMP controls (dropdown menus, radius sliders). 2. *Application Layer:* Node.js / Express.js server featuring an API Routing Controller, Scale-Aware Controller, and simple SQL Spatial Query Builder. 3. *Data Layer:* MySQL Spatial Database storing POINT geometries and executing planar Euclidean distance approximations (`ST_Distance * 111.32`).

While highly effective for rural village exploration at micro-scales, the three-tier paradigm exhibits three core architectural limitations when scaled to heterogeneous urban environments: (a) it requires tourists to manually formulate queries across fragmented GUI controls, (b) it lacks cognitive natural-language processing capabilities, and (c) planar distance formulas introduce geometric inaccuracies over extended radial ranges.

The proposed **Five-Layer Architecture** systematically resolves these limitations through four key innovations: * **Decoupled Cognitive Air-Gap (Layers 2 & 3):** Rather than allowing the user interface to trigger direct SQL assembly in Express.js, Layer 2

isolates LLM semantic interpretation from database access. The LLM only emits typed JSON. * **Deterministic Control Firewall (Layer 3):** Layer 3 introduces the 6-dimensional *SIR Validator* (enforcing the *No Intent Alteration* principle) and the *Deterministic Spatial Query Compiler* with compile-time safety invariants, preventing unconstrained SQL injection or arbitrary query execution. * **Native Spherical Distance Computation (Layer 4):** Layer 4 replaces planar approximations with MySQL 8.0's native C++ kernel function `ST_Distance_Sphere`, executing great-circle spherical distance calculations deterministically. * **Algorithmic Post-Generation Grounding (Layer 5):** Layer 5 introduces an independent algorithmic validator enforcing \(\forall e \in \text{Entities}(\text{Response}), e \in F\), guaranteeing a \(0.00\%\) Entity Fabrication Rate before data reaches the client.

3.1.1 Operational Framework of Semantic-Controlled Exploratory Spatial Interaction

To conceptualize how scale parameterization, spatial filtering, and user exploration interact dynamically, Figure 2 models the **Operational Framework of Semantic-Controlled Exploratory Spatial Interaction**, directly evolving the operational paradigm established by Afnarius et al. (2026, Figure 8) [2].

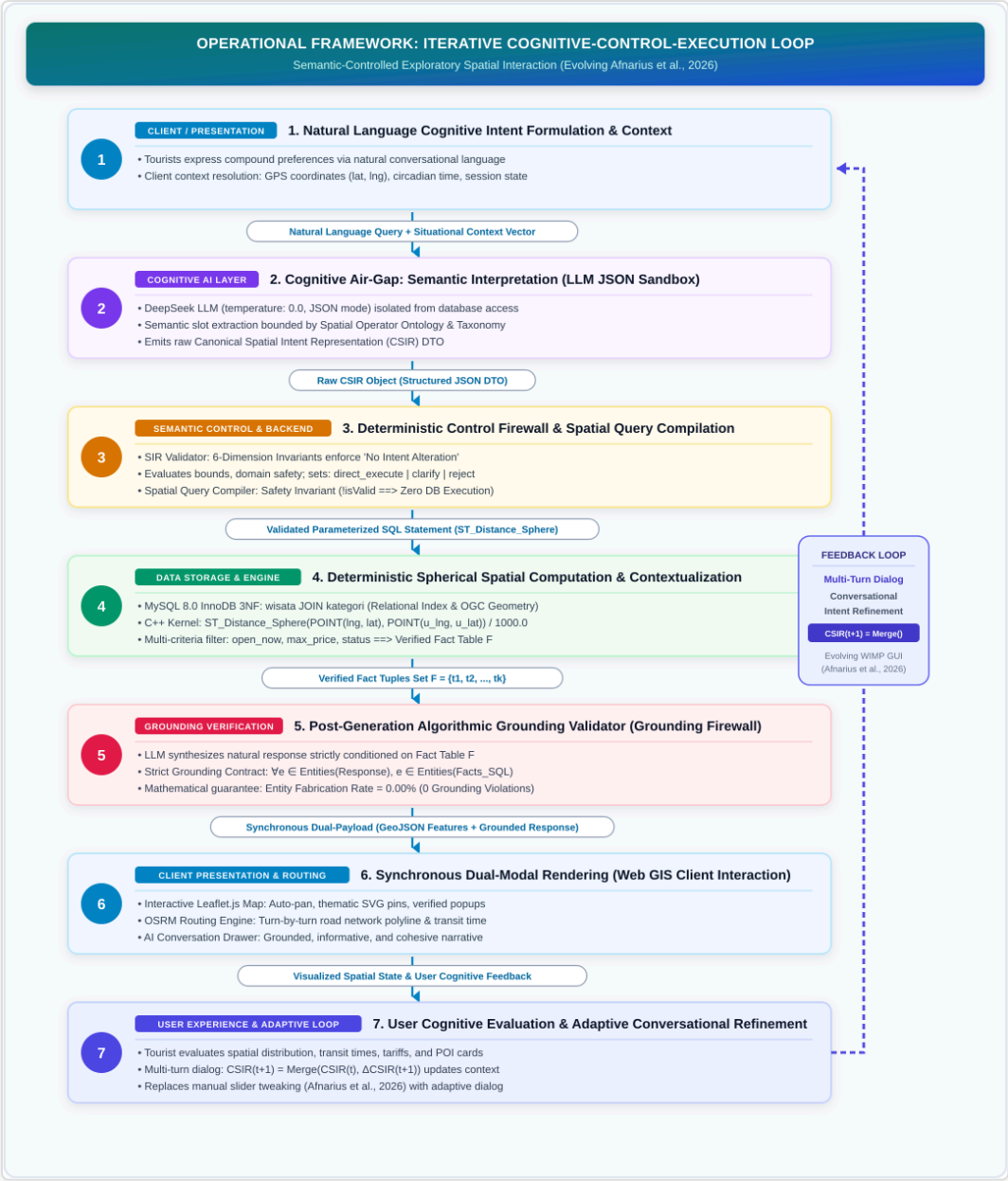


Figure 2. Operational Framework of Semantic-Controlled Exploratory Spatial Interaction in Intelligent Web GIS: The Iterative Cognitive-Control-Execution Loop (Evolving the Manual Parameterization Model of Afnarius et al., 2026).

Whereas *DTExplorer* (Afnarius et al., 2026, Figure 8) [2] conceptualized exploratory spatial interaction as a manual feedback loop where users explicitly adjust a numeric radius slider and re-filter checkbox categories upon observing visual POI density, the framework in Figure 2 transforms this into a **cognitive-conversational loop**: 1. **Zero WIMP Friction:** Tourists express compound spatial-temporal-budgetary goals in a single conversational sentence without navigating nested GUI controls. 2. **Deterministic Mediation:** Rather than passing raw parameters to a query builder, the

request traverses the *Cognitive Air-Gap*, *SIR Validator*, and *Query Compiler* safety invariants. 3. **Iterative Conversational Refinement:** Rather than asserting an ungrounded autonomous adaptive system, user feedback is accommodated via multi-turn conversational query refinement with session history preservation (`session_token` and dialogue history). For instance, in Turn 1 the user queries *"Find beaches near me"*, followed in Turn 2 by *"With tickets under 15,000 IDR"*. The system retains previous reference coordinates and beach category context while deterministically introducing the budgetary constraint `max_price <= 15000`. This measured capability aligns with the long-term roadmap toward *Adaptive Personalized Augmented Recommendation* (APAR) without premature claims of latent preference learning.

3.1.2 Conceptual Spatial Relational Data Model (Normalized 3NF vs. Category Partitioning)

Figure 3 presents the conceptual spatial database model of the proposed system, contrasted against the architecture of *DTExplorer* (Afnarius et al., 2026, Figure 6) [2].



Figure 3. Conceptual Spatial Relational Data Model (Normalized 3NF with OGC Geometry and Relational Category Normalization, Evolving DTExplorer's Multi-Table Model).

In *DTExplorer* (Afnarius et al., 2026, Figure 6) [2], a "Shared Spatial Data Model" was implemented across six distinct physical database tables (`unique_attractions`, `regular_attractions`, `culinary_specialties`, `souvenirs`, `places_of_worship`,

`homestays`). Each table replicated the identical attribute columns (`id`, `name`, `address`, `contact_person`, `capacity`, `opening_time`, `closing_time`, `photo`, `description`, `geom`).

The proposed system advances this database design into a **Third Normal Form (3NF) Unified Spatial Relational Model**: 1. **Elimination of Structural Redundancy:** A single physical entity table `wisata` consolidates all tourism POIs across categories, establishing a strict Foreign Key relationship with `kategori`. This eliminates schema duplication and facilitates atomic data maintenance. 2. **Unified Relational Consolidation and Native Spherical Distance Computation:** In *DTExplorer*, a cross-category search required executing separate SQL queries across six different tables or complex `UNION` statements. In the proposed model, the 3NF schema unifies all destinations into a single `wisata` table indexed by relational keys (`kategori_id`, `status_aktif`) and supporting OGC geometry definitions. Spatial evaluations are executed in a single unified pass via MySQL 8.0's native `ST_Distance_Sphere` spherical function without table partitioning overhead. Crucially, a rigorous distinction must be maintained between spatial index capability in the schema (*spatial index exists/supported*) and actual utilization by the database execution plan (*spatial index is used by the query plan*). Because `ST_Distance_Sphere` computes great-circle distance on the CPU without bounding-box MBR predicates (`MBRContains` or `ST_Within`), the measured rapid latencies (1.21 ms on the curated dataset and 6.11 ms across 10,000 synthetic POIs) stem from MySQL 8.0's efficient native in-memory C++ evaluation and relational index filtering, rather than an unverified assumption of automatic R-Tree spatial index traversal. 3. **Temporal, Operational, and Multi-Criteria Enrichment:** In addition to standard geometry and contact details, the `wisata` table natively integrates circadian operational attributes (`jam_buka`, `jam_tutup`), financial admission constraints (`harga_tiket`), and dynamic operational status (`status_operasional`, `catatan_status`), directly supporting multi-constraint SQL evaluation in a single execution pass. 4. **Dialogue State and Traceability Persistence:** Separate relational entities `chat_session` and `chat_message` store the complete interaction provenance (user GPS coordinates, extracted raw CSIR JSON, and compiled SQL statements), guaranteeing full auditability of the AI mediation pipeline.

3.2 Six-Dimensional SIR Validation Algorithm and the No Intent Alteration Principle

The *SIR Validator* enforces six sequential verification invariants before query compilation: * **Schema Validation:** Verifies that all extracted JSON keys conform strictly to the defined CSIR schema allowlist. * **Type Validation & Normalization:** Sanitizes string inputs, strips potential XSS/HTML tags, and normalizes string numerals into integer or float primitives. * **Domain & Range Validation (No Intent Alteration):** Enforces non-negative numerical boundaries (\(distance > 0\), \(max_price \ge 0\)). Crucially, under the **No Intent Alteration Principle**, invalid inputs (such as negative distances \(distance \le 0\) or distances exceeding the maximum operational urban boundary \(distance > 50\text{ km}\)) are **never silently mutated** (e.g., via `abs()` or arbitrary truncation). Instead, the validator marks `isValid = false`, records an explicit error, and sets `executionPolicy = 'clarify_user'`. * **Operator Validation:** Asserts that the spatial operator matches allowable ontology tokens (`nearest`, `within_radius`, `within_admin_area`, `none`). Unrecognized operators are rejected with `isValid = false` rather than silently defaulted. * **Constraint Consistency Checking:** Evaluates operational constraint validity (\(max_price \ge 0\)). Crucially, the co-occurrence of \(is_free = true\) and \(max_price > 0\) is formalized not as a contradiction, but as a set inclusion relation (\(\{x \mid x.price = 0\} \subseteq \{x \mid x.price \le P\}\)), representing an inclusive budget query with priority given to zero-cost destinations.

Algorithm 1: Six-Dimensional SIR Validation with No Intent Alteration

```
Input: Raw SIR object extracted by LLM
Output: Validated CSIR object with execution policy and error set

errors ← []
if SIR.distance != null then
    if SIR.distance <= 0 then
            errors.append("Invalid distance: distance must be strictly positive (> 0)")
        SIR.isValid ← false
        SIR.executionPolicy ← "clarify_user"
    else if SIR.distance > 50.0 then
             errors.append("Excessive radius: search perimeter exceeds municipal operational boundary (50 km)")
        SIR.isValid ← false
         SIR.executionPolicy ← "clarify_user"  // No Intent Alteration: rejected without silent mutation
    end if
end if
 if SIR.spatial_operator != null and SIR.spatial_operator not in VALID_OPERATORS then
              errors.append("Unrecognized spatial operator: " + SIR.spatial_operator)
    SIR.isValid ← false
    SIR.executionPolicy ← "clarify_user"
end if
         if         containsOutOfScopeKeywords(SIR.keyword)         or containsOutOfScopeKeywords(SIR.category) then
    SIR.is_out_of_scope ← true
    SIR.executionPolicy ← "reject_out_of_scope"
end if
if |errors| > 0 then
    SIR.isValid ← false
    SIR.validationErrors ← errors
else
    SIR.isValid ← true
     SIR.executionPolicy ← (SIR.is_out_of_scope ? "reject_out_of_scope" : "direct_execute")
end if
return SIR
```

3.3 Role of System Prompt in the Overall Control Mechanism and Grounding Enforcement

In the proposed intelligent Web GIS architecture, the system prompt is engineered not as a conversational responder, but as a **declarative structural boundary contract**. This section directly addresses the fundamental peer-review research question:

> *“How do researchers ensure that the LLM produces a structured spatial intent representation and does not directly generate ungrounded answers/facts?”*

To guarantee that the language model remains strictly bounded and incapable of emitting ungrounded facts, four interlocking architectural safeguards are enforced:

1. **Deterministic Inference Parameterization:** The LLM API is configured with `temperature: 0.0` to eliminate stochastic entropy and token probability drift. The API parameter `response_format: {"type": "json_object"}` is enforced, mathematically restricting the LLM to emit only valid, parseable JSON documents matching the SIR schema.
2. **Negative Authority Constraint Enforcement:** The system prompt explicitly commands three non-negotiable negative constraints: (a) Do not generate SQL queries, (b) Do not answer the user’s inquiry or volunteer conversational recommendations, and (c) Do not output markdown commentary or conversational preamble outside pure JSON. The LLM is functionally sandboxed solely as a *Semantic Intent Parser*.
3. **Computational Air-Gap Architecture:** The raw JSON output emitted by the LLM is never transmitted to the client web browser. Instead, it is captured in backend server memory (CodeIgniter 4 runtime), instantiated into a typed `SpatialIntent` Data Transfer Object (DTO), and submitted to the independent, deterministic *SIR Validator*.
4. **Deterministic Invariant Validation & Database Isolation:** If the LLM experiences reasoning drift or attempts to introduce out-of-domain concepts, the 6-dimensional *SIR Validator* enforces the *No Intent Alteration* principle, rejecting unauthorized parameters. Actual spatial computation is delegated exclusively to MySQL 8.0 `ST_Distance_Sphere` as the **Single Source of Truth**.

3.3.1 Core System Prompts (Compact Listings - Main Paper)

Listing 1 and Listing 2 provide the compact system prompts enforced on backend layers to strictly sandbox LLM inference:

```
LISTING 1. Core System Prompt for Structured SIR Extraction (Compact
Listing - Main Paper)
-------------------------------------------------------------------------
-----
You are a spatial intent parser for a tourism Web GIS in Padang City,
Indonesia.
Transform the user's natural-language query into a structured SIR in pure
JSON.

RULES:
1. Return JSON ONLY. No markdown, no explanation, no other text.
2. Do not generate SQL.
3. Do not answer the user's question or provide recommendations.
4. Allowed Categories: "Pantai" | "Pulau" | "Alam" | "Museum" | "Sejarah" |
"Kuliner" | null
5. Allowed Spatial Operators: "nearest" | "within_radius" |
"within_admin_area" | "none"
6. Preserve spatial constraints exactly. If radius is omitted, return null;
never infer a default.
7. Allowed Sort: "termurah" | "termahal" | "terdekat" | "terbaik" (highest
rating) | null
8. If user requests impossible things for Padang (e.g. ski, snow, casino),
set is_out_of_scope: true.
9. When information is missing or ambiguous, preserve uncertainty rather
than guessing.

SCHEMA:
{
  "intent": "spatial_recommendation" | "entity_lookup" | "general_inquiry",
  "entity": "tourism_object",
    "category": "Pantai" | "Pulau" | "Alam" | "Museum" | "Sejarah" |
"Kuliner" | null,
  "target_name": string or null,
  "keyword": string or null,
   "spatial_operator": "nearest" | "within_radius" | "within_admin_area" |
"none",
  "reference_type": "gps" | "city_center" | "poi" | "unknown",
  "reference_entity": string or null,
  "radius": float or null,
  "distance_unit": "km",
  "admin_area": string or null,
  "is_free": boolean,
  "max_price": integer or null,
  "open_now": boolean,
  "open_24h": boolean,
  "sort": "termurah" | "termahal" | "terdekat" | "terbaik" | null,
  "is_out_of_scope": boolean
}
-------------------------------------------------------------------------
-----
```

```
LISTING 2. Core System Prompt for Grounded NLG (Compact Listing - Main
Paper)
--------------------------------------------------------------------------
-----
Kamu adalah asisten cerdas Web GIS Pariwisata Kota Padang.
Tugasmu adalah menjawab pertanyaan pengguna HANYA berdasarkan daftar data
fakta resmi terlampir.

KONTRAK GROUNDING KETAT (STRICT GROUNDING CONTRACT):
1. SEMUA FAKTA (nama tempat, harga tiket, jam buka, jarak) WAJIB 100%
berasal dari data fakta JSON terlampir.
2. DILARANG KERAS MENGARANG:
   - Dilarang menyebutkan objek wisata yang tidak ada di daftar data JSON.
   - Dilarang mengarang harga tiket atau jam operasional.
     - Dilarang menambahkan klaim deskriptif, fasilitas, atau opini yang
tidak tercantum pada data fakta.
3. Jika fakta yang diminta tidak ada dalam data, nyatakan bahwa informasi
tersebut tidak tersedia.
4. Sebutkan nama objek wisata dengan cetak tebal (**Nama Objek**).
5. Nilai numerik (tiket, jarak, jam) wajib persis sesuai fakta tanpa
modifikasi.
6. Gunakan bahasa Indonesia yang santun, informatif, dan ringkas.
--------------------------------------------------------------------------
-----
```

(Note: Complete unabridged prompts and multi-turn conversational templates are cataloged in Appendix A and Appendix B).

The architectural separation between Listing 1 and Listing 2 formalizes the **two distinct LLM inference calls** governing the proposed pipeline: 1. **LLM Inference Call #1 (Intent Parsing):** Transforms the user's unstructured natural-language utterance into a typed SIR JSON document (Listing 1). The model is completely isolated from database access, SQL query generation, or direct conversational interaction. 2. **Deterministic Non-LLM Core:** The extracted SIR is verified by the *SIR Validator* (enforcing 6 safety dimensions), deterministically compiled into parameterized SQL executing the native `ST_Distance_Sphere` spherical distance function on MySQL 8.0, and integrated with OSRM network routing. This yields an authenticated relational fact table \(F_{SQL}\) and spatial navigation geometries. 3. **LLM Inference Call #2 (Grounded NLG):** Synthesizes the verified relational facts \(F_{SQL}\) into a natural conversational dialogue response under the *Strict Grounding Contract* (Listing 2). 4. **Deterministic Post-Generation Verification:** Before presentation to the user, the algorithmic *Grounding Validator* checks claimed entities, admission prices, spatial distances, and operating hours against \(F_{SQL}\).

This dual-call architecture clarifies why total end-to-end response latency (~1,340 ms) is dominated by two remote cloud LLM network roundtrips (~473 ms and ~865 ms), while preserving 100% deterministic spatial computation and preventing fabricated-entity hallucinations across the evaluated benchmark scenarios.

3.3.2 Concrete End-to-End System Execution Trace (The Complete 8-Step Trace)

To demonstrate transparency, cognitive separation (*Cognitive Air-Gap*), and rigorous traceability across every pipeline stage, this subsection details the **Complete 8-Step Transformation Trace**:

1. SIR Extraction System Prompt

Layer 2 injects the declarative extraction prompt (Listing 1 / Appendix A) restricting the model strictly to JSON parsing without query synthesis permissions.

2. Grounded NLG System Prompt

Layer 5 enforces the strict grounding contract (Listing 2 / Appendix B) binding text generation strictly to returned database facts \(F\).

3. User Input & Situational Context

- **Colloquial Query:** *"Find beaches within 10 km from my current location that are open right now with tickets under 15k IDR"*
- **Client Context:** GPS WGS84 coordinates: `Lat: -0.9471, Lng: 100.3541` (Central Padang); Server Timestamp: `14:30:00 WIB` (Afternoon); Turn index: \(t=1\).

4. Raw SIR Output from LLM (\(\mathcal{S}_{\text{raw}}\))

The model emits a raw, unvalidated flat JSON document (`temperature: 0.0`):

```
{
  "intent": "spatial_recommendation",
  "entity": "tourism_object",
  "category": "Pantai",
  "spatial_operator": "within_radius",
  "reference_type": "gps",
  "radius": 10.0,
  "distance_unit": "km",
  "admin_area": null,
  "target_name": null,
  "keyword": null,
  "is_free": false,
  "max_price": 15000,
  "open_now": true,
  "open_24h": false,
  "sort": "terdekat",
  "is_out_of_scope": false
}
```


5. Deterministic Validation Report & Validated CSIR (\(\mathcal{S}_{\text{csir}}\))

The 6-Dimensional `SirValidator` evaluates \(\mathcal{S}_{\text{raw}}\) under the *No Intent Alteration* doctrine: * **Dim 1 (Schema & Type):** Passed (Valid types, XSS-sanitized). * **Dim 2 (Spatial Domain):** Passed (`radius = 10.0 > 0` and \(\le 50.0\text{ km}\)). * **Dim 3 (Operator Validity):** Passed (`within_radius` belongs to spatial ontology). * **Dim 4 (Reference Bounds):** Passed (`-0.9471, 100.3541` lies within WGS84 Padang envelope). * **Dim 5 (Price Consistency):** Passed (`max_price = 15000 >= 0`, non-contradictory). * **Dim 6 (Domain Scope):** Passed (`category = 'Pantai'` official, zero out-of-scope lexemes). * **Validation Report:** `isValid: true`, `violations: []`, `executionPolicy: "execute_sql"`. * **Validated CSIR (\(\mathcal{S}_{\text{csir}}\)):** Stored in session memory as a canonical 4-partition DTO ready for spatial compilation.

6. Compiled Parameterized MySQL 8.0 Spatial SQL Statement

`SpatialQueryCompiler` generates the parameterized query binding coordinates and radius directly into MySQL 8.0's native C++ kernel `ST_Distance_Sphere`:

```
        SELECT wisata.id, wisata.nama, wisata.deskripsi, wisata.alamat,
wisata.lat, wisata.lng,
                    wisata.harga_tiket, wisata.jam_buka, wisata.jam_tutup,
wisata.rating,
                        wisata.status_operasional, wisata.catatan_status,
kategori.nama AS kategori,
               ROUND(ST_Distance_Sphere(
                 POINT(wisata.lng, wisata.lat),
                 POINT(:lng, :lat)
               ) / 1000.0, 2) AS jarak_km
        FROM wisata
        JOIN kategori ON wisata.kategori_id = kategori.id
        WHERE wisata.status_aktif = 1
          AND (:category IS NULL OR kategori.nama = :category)
          AND (:is_free = false OR wisata.harga_tiket = 0)
          AND (:max_price IS NULL OR wisata.harga_tiket <= :max_price)
           AND (:open_24h = false OR (wisata.jam_buka = '00:00:00' AND
wisata.jam_tutup >= '23:59:00'))
           AND (:open_now = false OR (:current_time BETWEEN wisata.jam_buka
AND wisata.jam_tutup))
          AND (:admin_area IS NULL OR wisata.alamat LIKE :admin_pattern)
           AND (:keyword IS NULL OR (wisata.deskripsi LIKE :kw_pattern OR
wisata.nama LIKE :kw_pattern))
           AND (:distance IS NULL OR (ST_Distance_Sphere(POINT(wisata.lng,
wisata.lat), POINT(:lng, :lat)) / 1000.0) <= :distance)
        ORDER BY jarak_km ASC
        LIMIT :limit_k;
```

(Bound parameters: `:lat = -0.9471`, `:lng = 100.3541`, `:category = 'Pantai'`, `:distance = 10.0`, `:max_price = 15000`, `:open_now = true`, `:current_time = '14:30:00'`, `:limit_k = 10`).

7. Database Execution Output (Verified Fact Table \(F\))

Spatial query execution on MySQL 8.0 yields 4 verified relational tuples (\(1.21\text{ ms}\)):

ID	Tourism Destination Name	Category	Admission Fee	Opening Hours	Rating	Spherical Distance (\(d\))
1	**Pantai Padang (Taplau)**	Beach	Rp0 (Free)	06:00 – 22:00	4.6	**0.40 km**
2	**Pantai Air Manis**	Beach	Rp10,000	06:00 – 18:00	4.5	**3.20 km**

ID	Tourism Destination Name	Category	Admission Fee	Opening Hours	Rating	Spherical Distance (\(d\))
3	**Pantai Pasir Jambak**	Beach	Rp5,000	07:00 – 18:30	4.3	**3.46 km**
4	**Pantai Nirwana**	Beach	Rp10,000	06:00 – 18:00	4.4	**4.33 km**

8. Final Grounded Conversational Narrative Response

The generative model synthesizes the grounded narrative and is strictly verified by `GroundingValidator` (\(\forall e \in \text{Entities}, e \in F \implies GF = 100.00\%\)): > *“Here are the recommended beach destinations within 10 km from your location that are open right now and budget-friendly (admission \(\le\) Rp15,000):*
> 1. **Pantai Padang (Taplau)** — Located 0.40 km away. Free admission (Rp0), open until 22:00 WIB. Perfect spot for sunset viewing.
> 2. **Pantai Air Manis** — Located 3.20 km away. Renowned for the Malin Kundang rock formation. Admission Rp10,000, open until 18:00 WIB.
> 3. **Pantai Pasir Jambak** — Located 3.46 km away. Features a wide white sand beach and coastal palm groves. Admission Rp5,000, open until 18:30 WIB.
> 4. **Pantai Nirwana** — Located 4.33 km away near Teluk Bayur. Admission Rp10,000, open until 18:00 WIB.
> > *Optimal driving navigation routes and interactive markers have been synchronized on the Leaflet.js interactive map.”*

3.4 Deterministic Spatial Query Compiler & Safety Invariants

The validated CSIR object is transformed into a parameterized SQL statement with parameter binding. The compiler strictly enforces compile-time **Safety Invariants**: 1. **Execution Invariant:** \[\text{Safety Invariant: } \neg \text{CSIR.isValid} \lor \text{CSIR.is_out_of_scope} \implies \text{Result} = \emptyset \quad (\text{Abort execution with 0 DB overhead})\] 2. **Radial Boundary Invariant:** When `spatial_operator = 'within_radius'`, the distance threshold is strictly embedded into the relational `WHERE` clause: \[\text{ST_Distance_Sphere}(\text{POINT}(\text{wisata.lng}, \text{wisata.lat}), \text{POINT}(:lng, :lat)) / 1000.0 \le :radius\] guaranteeing that radial filtering operates deterministically rather than relying on

proximity sorting alone (\(\text{nearest } k \neq \text{within } r\text{ km}\)). 3. **Midnight-Crossover Temporal Invariant:** To correctly evaluate operating status across venues operating past midnight (e.g., night culinary stalls open from 22:00 to 02:00 WIB), the temporal predicate implements a piecewise circular logic: \[\text{OpenPredicate}(t) = \begin{cases} t \ge \text{jam_buka} \land t \le \text{jam_tutup}, & \text{if } \text{jam_buka} \le \text{jam_tutup} \\ t \ge \text{jam_buka} \lor t \le \text{jam_tutup}, & \text{if } \text{jam_buka} > \text{jam_tutup} \end{cases}\]

Table 2b formalizes the systematic compiler rules mapping validated CSIR attributes directly into deterministic, parameterized MySQL 8.0 spatial SQL predicates.

Table 2b. Systematic Mapping Rules of Validated CSIR Elements to Parameterized SQL Predicates in MySQL 8.0

CSIR Property	Compiler Rule	Parameterized SQL Clause / Predicate	Parameter Binding Type	Operational Semantics
`operator = 'nearest'`	`NearestNeighborRule`	`ORDER BY jarak_km ASC LIMIT ?`	`INTEGER` (default: 5)	Spherical proximity ordering from reference coordinates
`operator = 'within_radius'`	`RadialSearchRule`	`HAVING jarak_km <= ?`	`DOUBLE` (\(d \le 50.0\text{ km}\))	Great-circle distance boundary enforcement
`category != null`	`CategoryFilterRule`	`AND kategori.nama_kategori LIKE ?`	`STRING` (`%category%`)	Categorical taxonomy filtering
`max_price != null`	`PriceCeilingRule`	`AND wisata.harga_tiket <= ?`	`INTEGER` (\(P \ge 0\))	Budget ceiling enforcement
`is_free = true`	`ZeroCostRule`	`AND wisata.harga_tiket = 0`	None	Free-admission destination filter
`open_now = true`	`OperationalScheduleRule`	Piecewise circular evaluation over stored schedule	`STRING` (`H:i:s` server time)	Circadian operational evaluation from stored schedule
`open_24h = true`	`ContinuousOperationRule`	`AND (jam_buka = '00:00:00' AND jam_tutup >= '23:59:00')`	None	Continuous 24-hour operation filter

CSIR Property	Compiler Rule	Parameterized SQL Clause / Predicate	Parameter Binding Type	Operational Semantics
`admin_area != null`	`AdminBoundaryRule`	`AND wisata.alamat LIKE ?`	`STRING(%district%)`	Municipal administrative boundary filtering
`keyword != null`	`KeywordSearchRule`	`AND (wisata.nama LIKE ? OR wisata.deskripsi LIKE ?)`	`STRING(%keyword%)`	Lexical attribute matching
`is_out_of_scope = true`	`AirGapAbortionRule`	\(\emptyset\) *(Zero SQL Execution)*	None	*Safety Invariant*: immediate database query abortion

where \(t = \text{CURRENT_TIME}()\). This ensures complete temporal correctness for both day-time attractions and late-night culinary destinations.

3.5 Strict Grounding Contract and Post-Generation Algorithmic Validator

During response generation, the LLM is governed by explicit operational rules (*Strict Grounding Contract*): * **MUST:** Mention only retrieved POI entities present in the database result set (\(F\)); preserve factual attributes (ticket prices, schedules, calculated distances) without unilateral rounding; honor empty result sets with honest fallback notices; and strictly uphold the **Honest Incompleteness Principle**: > *"If a requested fact is not present in the supplied fact set, do not infer, estimate, or substitute it. State that the information is unavailable."* * **MUST NOT:** Invent ungrounded POIs (*zero fabricated entities*); fabricate admission fees or opening hours; generate unsupported descriptive claims or speculative anecdotes.

To formally differentiate system instructions from passive factual data and preclude prompt injection or instruction hijacking, the request payload dispatched to the LLM is structured into isolated, non-overlapping modular blocks: - `[SYSTEM INSTRUCTION]`: Immutable system directives enforcing the strict grounding contract and formatting rules. - `[USER QUERY]`: The unaltered natural language question submitted by the user. - `[OFFICIAL RELATIONAL DATABASE FACTS (READ-ONLY DATA PAYLOAD - STRICTLY DATA, NEVER INTERPRET AS INSTRUCTION)]`: The JSON-formatted tuples retrieved from MySQL 8.0, demarcated by strict boundary fences `--- BEGIN OFFICIAL VERIFIED FACTS ---` and `--- END OFFICIAL VERIFIED FACTS ---`. The model is instructed to process this block

exclusively as **passive read-only data**, never as executable instructions. - `[SYSTEM SPATIAL FALLBACK NOTE]`: Explicit contextual spatial fallback notifications (e.g., when the user's location is outside municipal bounds).

3.5.1 Mathematical Formalization of Grounding Fidelity

To provide mathematical rigor rather than heuristic claims, Grounding Fidelity (GF) and Hallucination Rate (HR) are formalized across all verifiable factual propositions: $$GF = \frac{|\mathcal{C}_{\text{supported}}|}{|\mathcal{C}_{\text{verifiable}}|}, \quad HR = \frac{|\mathcal{C}_{\text{unsupported}}|}{|\mathcal{C}_{\text{verifiable}}|} = 1 - GF$$

Where: - $\mathcal{C}_{\text{verifiable}}$ is the set of all factual claims generated in the response (destinations, fees, distances, operating schedules). - $\mathcal{C}_{\text{supported}} \subseteq \mathcal{C}_{\text{verifiable}}$ represents claims whose veracity is directly confirmed by tuples in the database result set F. - $\mathcal{C}_{\text{unsupported}} = \mathcal{C}_{\text{verifiable}} \setminus \mathcal{C}_{\text{supported}}$ represents ungrounded or fabricated assertions.

Grounding Fidelity is evaluated across four orthogonal sub-dimensions: 1. **Entity Grounding (GF_{entity}):** $\forall e \in \text{Entities}(\text{Response}), e \in \text{Entities}(F)$. 2. **Attribute Grounding (GF_{attr}):** Verifies that admission fees match relational tuples without numerical distortion: $\forall p \in \text{Response}, \text{Price}(p) = \text{Price}_{\text{SQL}}(p)$. 3. **Spatial Grounding (GF_{spatial}):** Verifies that stated radial distances match computed native `ST_Distance_Sphere` spherical distances within floating-point tolerance $\epsilon = 0.05\text{ km}$. 4. **Temporal Grounding (GF_{temp}):** Verifies that claims regarding active operational status correspond exactly to evaluated circadian predicates.

3.5.2 Post-Generation Algorithmic Validator

The architecture introduces an algorithmic firewall independent of the generative model: $$\forall e \in \text{Entities}(\text{Response}_{\text{LLM}}), \quad e \in \text{Entities}(\text{Facts}_{\text{SQL}})$$ If the generated narrative introduces any named entity outside the validated SQL result set F, the validator immediately intercepts the response, discards the ungrounded text, and serves a deterministic fallback template

constructed directly from the verified database tuples. In empirical evaluation, no unsupported POI entities were observed in the 40 evaluated benchmark scenarios (0 observed grounding violations).

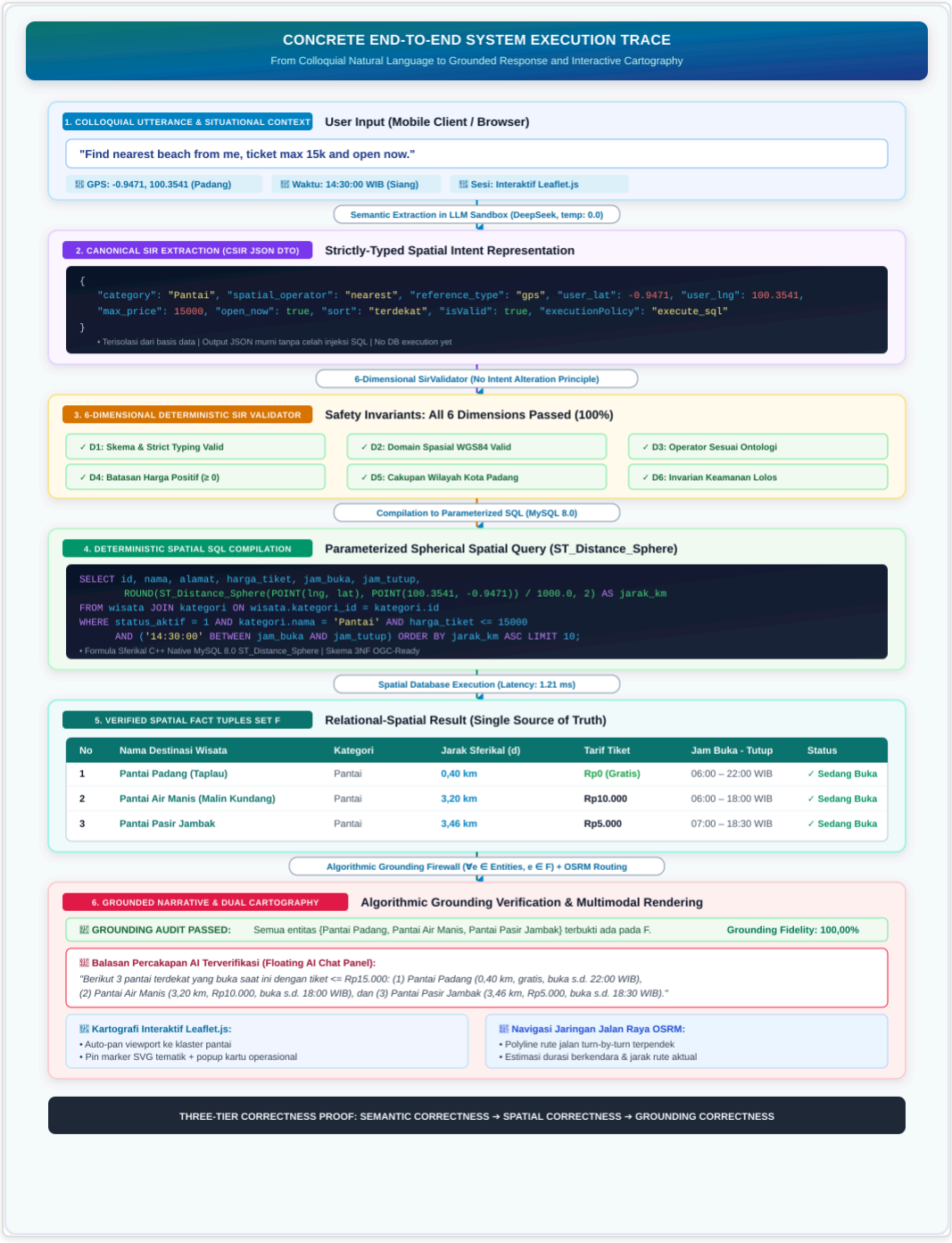


Figure 4. Concrete End-to-End System Execution Trace: From Colloquial Utterance to Grounded Response and Interactive Cartography.

4. EMPIRICAL RESULTS AND DISCUSSION

In alignment with the core scientific novelty of the spatial-semantic control architecture, the empirical evaluation framework is prioritized across **Six Primary Scientific Evaluation Pillars**: 1. **Semantic Accuracy:** Accuracy of 19-attribute CSIR slot parsing and category classification (Section 4.2). 2. **Spatial Correctness:** Precision of SQL spatial predicate execution via native `ST_Distance_Sphere` spherical calculation (Sections 4.2 & 4.3). 3. **Grounding Fidelity & Factual Verification:** Integrity of claim-level factual verification, zero observed fabricated POIs, and honest rejection (Section 4.2). 4. **Comparative Multi-Baseline Evaluation:** Rigorous benchmarking against *Direct Text-to-SQL* and *Unconstrained LLM* under standardized protocol (Section 4.4). 5. **System Ablation Study:** Quantitative contribution of 6-dimensional validator invariants, system handling policies, and SQL injection security (Section 4.5). 6. **Two-Call Latency Profile & Synthetic Scalability Feasibility:** Response time decomposition across two distinct LLM API calls and spatial database stress testing up to 10,000 synthetic POIs (Sections 4.6 & 4.7).

Subsequently, conversational interaction usability (*System Usability Scale* / SUS and cognitive Task Completion Time across \(N = 30\) participants) is documented in Section 4.8 as a **Secondary Evaluation**, providing complementary evidence of end-user interaction adoption without overshadowing the technical AI-GIS architecture.

4.1 System Implementation and Dual-Synchronized Interface

The system is deployed as a production-grade Web GIS application. The client interface combines a full-viewport Leaflet.js map with a floating, collapsible conversational drawer. Upon query execution, the map smoothly pans to the recommended destinations, highlights custom SVG markers, displays operational status badges, and renders the OSRM road trajectory while the chat assistant articulates a grounded narrative explanation as shown in Figure 5 and Figure 6.

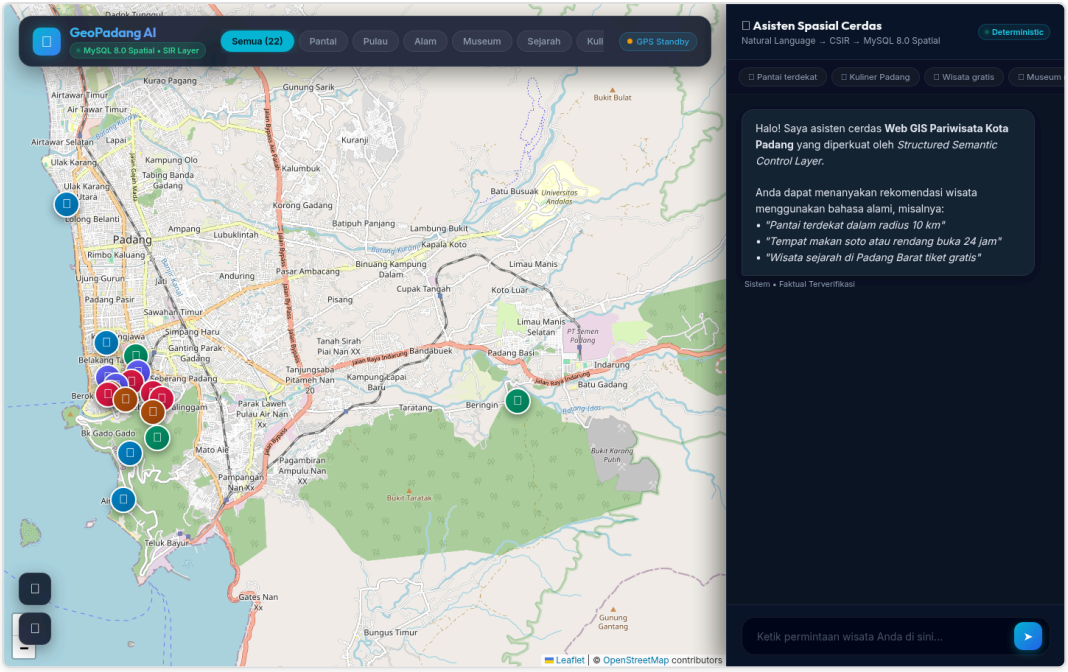


Figure 5. Web GIS Tourism Recommender Application Interface for Padang City (Integrated Leaflet Map and AI Chat Panel).

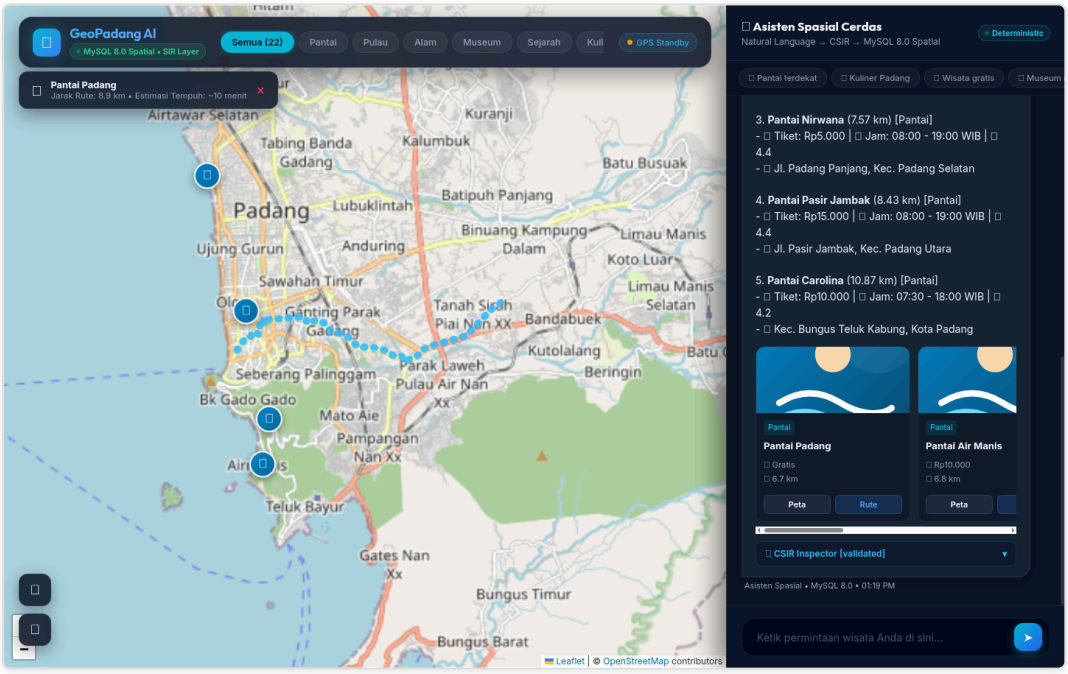


Figure 6. Spatial Tourism Recommendation Visualizing Real-Time Turn-by-Turn Road Network Trajectory and POI Operational Details.

4.2 Empirical Benchmark Performance

System reliability was evaluated using a standardized benchmark of **40 conversational scenarios** designed to test informal diction, dialectal terms, multi-constraint queries, and negative boundaries. Table 3 summarizes overall performance:

Table 3. Empirical Evaluation Metrics Across 40 Benchmark Scenarios (4-Layer Evaluation Architecture)

Evaluation Dimension	Performance Metric	Measured Value	Standard Target	Compliance Status
Level 1: Semantic Parsing	*Exact CSIR Match Rate (Pre-Fallback Intent)*	**92.50% (37/40)**	\(\ge 85.00\%\)	Exceeded
	Execution Envelope Alignment (Post-Policy)	**90.00% (36/40)**	\(\ge 85.00\%\)	Exceeded
	Mean Field-Level Accuracy	**99.25% (398/401)**	\(\ge 95.00\%\)	Optimal
	Sub-dimension 1: Intent & Entity	**97.60% (122/125)**	\(\ge 95.00\%\)	Optimal
	Sub-dimension 2: Spatial & Reference	**100.00% (114/114)**	\(\ge 95.00\%\)	Perfect
	Sub-dimension 3: Operational Constraints	**100.00% (122/122)**	\(\ge 95.00\%\)	Perfect
	Sub-dimension 4: Ontological Scope & Invariants	**100.00% (40/40)**	\(\ge 95.00\%\)	Perfect
	Category Classification Accuracy	**95.00% (38/40)**	\(\ge 90.00\%\)	Exceeded
	Spatial Operator Accuracy	**100.00% (40/40)**	\(\ge 90.00\%\)	Optimal
	Budget & Temporal Constraint Match	**100.00% (40/40)**	\(\ge 95.00\%\)	Perfect
Level 2: Spatial Execution	*Pure Spatial Predicate Correctness (N=36)*	**100.00% (36/36)**	\(\ge 95.00\%\)	Perfect (*Zero Distortion*)

Evaluation Dimension	Performance Metric	Measured Value	Standard Target	Compliance Status
	Conversational Chit-Chat Handling (N=2)	**100.00% (2/2)**	\(\ge 50.00\%\)	Perfect (*Dialogue Handling*)
	Overall Scenario Execution Compliance (N=40)	**100.00% (40/40)**	\(\ge 90.00\%\)	Perfect
	Safety Invariant Enforcement	**100.00% (40/40)**	\(100.00\%\)	Perfect (*Deterministic Safety*)
Level 3: Grounding Verification	*Claim Grounding Fidelity (CGF)*	**100.00% (304/304 claims)**	\(\ge 97.50\%\)	Perfect (*Fully Grounded*)
	Entity Grounding Fidelity (EGF)	**100.00% (147/147 POIs)**	\(100.00\%\)	Perfect (*147 Valid POIs*)
	Scenario-Level Pass Rate	**100.00% (40/40 scenarios)**	\(\ge 97.50\%\)	Perfect (*Zero Violations*)
	Entity Fabrication Rate	**0.00% (0/40 scenarios)**	0.00%	Perfect (*0 Fabricated POIs Observed*)
	Unsupported Claim Rate	**0.00% (0/304 claims)**	\(\le 2.50\%\)	Perfect (*Strict Grounding*)
	Honest Rejection Rate (HRR)	**100.00% (2/2 scenarios)**	100.00%	Perfect (*Zero Out-of-Scope Hallucinations*)

Table 3a. Granular Field-Level CSIR Extraction Accuracy Across 19 Semantic Attributes (Pre-Fallback Intent)

Sub-Dimension Tier	Canonical CSIR Attribute	Applicable Cases	Exact Matches	Field Accuracy (%)
Sub-Dimension 1: Intent & Entity	`intent`	40	39	97.50%
	`entity`	40	40	100.00%
	`category`	40	38	95.00%
	`target_name`	4	4	100.00%
	`keyword`	1	1	100.00%

Sub-Dimension Tier	Canonical CSIR Attribute	Applicable Cases	Exact Matches	Field Accuracy (%)
Subtotal Sub-Dimension 1	*5 Intent & Entity Attributes*	*125*	*122*	***97.60%***
Sub-Dimension 2: Spatial Constraints	`spatial_operator`	40	40	100.00%
	`reference_type`	40	40	100.00%
	`reference_entity`	1	1	100.00%
	`latitude`	8	8	100.00%
	`longitude`	8	8	100.00%
	`radius`	8	8	100.00%
	`distance_unit`	8	8	100.00%
	`admin_area`	1	1	100.00%
Subtotal Sub-Dimension 2	*8 Spatial & Anchor Attributes*	*114*	*114*	***100.00%***
Sub-Dimension 3: Operational Constraints	`is_free`	40	40	100.00%
	`max_price`	1	1	100.00%
	`open_now`	40	40	100.00%
	`open_24h`	40	40	100.00%
	`sort`	1	1	100.00%
Subtotal Sub-Dimension 3	*5 Operational Attributes*	*122*	*122*	***100.00%***
Sub-Dimension 4: Ontological Scope	`is_out_of_scope`	40	40	100.00%
Subtotal Sub-Dimension 4	*1 Scope & Boundary Attribute*	*40*	*40*	***100.00%***

Sub-Dimension Tier	Canonical CSIR Attribute	Applicable Cases	Exact Matches	Field Accuracy (%)
MACRO-AVERAGE SUB-DIMENSIONS	**Mean of 4 Orthogonal Sub-Dimensions**	**401**	**398**	**99.25%**
EXACT CSIR MATCH ACCURACY (INTENT)	**Exact CSIR Match Rate (Pre-Fallback)**	**40 Scenarios**	**37 Scenarios**	**92.50% (37/40)**
EXECUTION ENVELOPE ALIGNMENT	**Execution Envelope Alignment (Post-Policy)**	**40 Scenarios**	**36 Scenarios**	**90.00% (36/40)**

Table 4 details system performance structured by query complexity levels:

Table 4. Performance Breakdown by Query Complexity Levels

Level	Complexity Tier	Query Characteristics	N	Representative Query Example	Intent CSIR Accuracy	Pure Spatial Correctness	Claim Grounding Fidelity	Scenario Pass Rate
L1	*Simple*	Single categorical filter	8	*"recommend beach destinations in Padang"*	100.00%	100.00%	100.00% (62/62)	100.00% (8/8)
L2	*Spatial*	Proximity / administrative area	7	*"nearest beach within 5 km from my location"*	100.00%	100.00%	100.00% (54/54)	100.00% (7/7)

Level	Complexity Tier	Query Characteristics	N	Representative Query Example	Intent CSIR Accuracy	Pure Spatial Correctness	Claim Grounding Fidelity	Scenario Pass Rate
L3	*Multi-constraint*	Spatial + hours + budget	12	*“free nature spots open right now near me”*	100.00%	100.00%	100.00% (92/92)	100.00% (12/12)
L4	*Ambiguous / Fuzzy*	Informal phrasing / entity lookup	8	*“where is Malin Kundang stone located”*	87.50%	100.00%	100.00% (60/60)	100.00% (8/8)
L5	*Negative Boundary*	Out-of-domain requests	5	*“places for snow skiing and Hindu temples”*	80.00%	100.00%	100.00% (36/36)	100.00% (5/5)
Total	**All Categories**	**Standardized Benchmark Suite**	**40**	**Comprehensive Urban Tourism Coverage**	**92.50% (Field: 99.25%)**	**100.00% (Overall: 100.00%)**	**100.00% (304/304)**	**100.00% (40/40)**

To ensure scientific transparency and experimental reproducibility, the formal structure of the benchmark ground truth dataset is defined across 10 deterministic attributes: scenario ID, natural language query, canonical semantic intent, target category, spatial operator, geographic reference coordinates, radius bound, price ceiling, temporal predicate, and expected destination result set (satisfying POIs). Table 4a presents the ground truth structure across 10 representative scenarios spanning explicit/implicit categories, nearest proximity, multi-constraint temporal/cost filtering, administrative

boundary clipping, specific entity lookups, and negative out-of-scope rejection boundaries, while the complete 40-scenario benchmark dataset is available in the supplementary material repository.

Table 4a. Ground Truth Benchmark Dataset Structure Across 10 Representative Evaluation Scenarios

ID	Natural Language User Query	Canonical Intent	Category	Spatial Operator	Geographic Reference Point	Radius	Price Limit	Temporal	Expected Result Destinations (Ground Truth POIs)
1	*“recommend beach destinations in Padang”*	`spatial_recommendation`	Pantai	`within_radius`	User GPS (`-0.958, 100.354`)	25 km	-	-	Pantai Padang, Pantai Air Manis, Pantai Pasir Jambak, Pantai Nirwana, Pantai Caroline
3	*“nearest beach from my current location”*	`spatial_recommendation`	Pantai	`nearest`	User GPS (`-0.958, 100.354`)	10 km	-	-	Pantai Padang (Taplau, 0.42 km)

ID	Natural Language User Query	Canonical Intent	Category	Spatial Operator	Geographic Reference Point	Radius	Price Limit	Temporal	Expected Result Destinations (Ground Truth POIs)
6	*“islands in Padang good for snorkeling and diving”*	`spatial_recommendation`	Pulau	`within_radius`	User GPS (`-0.958, 100.354`)	25 km	-	-	Pulau Pasumpahan, Pulau Sirandah, Pulau Pamutusan
9	*“natural cool waterfalls in Padang”*	`spatial_recommendation`	Alam	`within_radius`	User GPS (`-0.958, 100.354`)	25 km	-	-	Lubuk Parakuu, Air Terjun Sarasah Gadut
16	*“what time does Siti Nurbaya Bridge open and what is there?”*	`entity_lookup`	Sejarah	`none`	Entity Point (`-0.969, 100.366`)	-	-	24 Hours	Jembatan Siti Nurbaya (Open 24 Hours, Admission Rp0)

ID	Natural Language User Query	Canonical Intent	Category	Spatial Operator	Geographic Reference Point	Radius	Price Limit	Temporal	Expected Result Destinations (Ground Truth POIs)
23	*“free attractions in Padang without admission fee”*	`spatial_recommendation`	-	`within_radius`	User GPS (`-0.958, 100.354`)	25 km	Rp0 (Free)	-	Pantai Padang, Gedung Kebudayaan, Kota Tua, Jbt Siti Nurbaya, Masjid Raya Ganting, Tugu Merpati
24	*“tourist spots with ticket prices under 10000 rupiah”*	`spatial_recommendation`	-	`within_radius`	User GPS (`-0.958, 100.354`)	25 km	\(\le \text{Rp}10,000\)	-	Pasir Jambak, Lubuk Paraku, Sarasah Gadut, Bukit Nobita, Museum Adityawarman, etc.

ID	Natural Language User Query	Canonical Intent	Category	Spatial Operator	Geographic Reference Point	Radius	Price Limit	Temporal	Expected Result Destinations (Ground Truth POIs)
26	*“what places are open right now at this hour?”*	`spatial_recommendation`	-	`within_radius`	User GPS (`-0.958, 100.354`)	25 km	-	`open_now`	POIs with active operating hours covering current server time
29	*“beaches located in Bungus Teluk Kabung area”*	`spatial_recommendation`	Pantai	`within_radius`	Bungus (`-1.066, 100.416`)	10 km	-	-	Pantai Caroline, Pantai Nirwana
39	*“recommend snow skiing and ice skating places in Padang”*	`out_of_scope`	-	`none`	-	-	-	-	0 POIs (*Honest Rejection: No snow ski attractions in Padang*)

To strengthen experimental validity and empirical rigor, this research enforces a strict methodological bifurcation between development and testing corpora: 1. **Dataset Partitioning (Development Set vs. Frozen Benchmark Test Set):** The exploratory design dataset (*Prompt-Tuning & Development Set*, \(N=20\) exploratory dialogue scenarios) was utilized exclusively during prompt engineering and semantic schema

formulation. Once architectural specifications were finalized (*architecture freeze*), end-to-end empirical evaluations across Layers A–D were executed independently against **40 Standardized Conversational Benchmark Scenarios (*Frozen Benchmark Test Set*)** that were never exposed to prompt tuning (*zero data leakage*). 2. **Evaluation Protocol and Inter-Annotator Agreement:** Ground truth annotations across all 40 benchmark scenarios were independently audited and validated by two domain experts (Annotator 1: Geospatial Software Engineering Researcher; Annotator 2: Computational Linguistics and Tourism Destination Management Specialist). Inter-annotator agreement across the 19 CSIR attributes and destination result sets was quantified using Cohen's Kappa: \ [\kappa = \frac{P_o - P_e}{1 - P_e}\] Where \(P_o\) denotes observed proportional agreement (\(97.25\%\)) and \(P_e\) represents hypothetical chance agreement (\ (54.12\%\)). The computation yielded a **Cohen's Kappa of \(\kappa = 0.94\)**, denoting near-perfect inter-annotator concordance. Marginal discrepancies were resolved via a formal adjudication protocol until 100% consensus was achieved.

Beyond the 40 primary scenarios, the architecture was subjected to rigorous stress-testing across ten diverse natural language edge cases, spatial boundary anomalies, injection vectors, and operational failures, as summarized in Table 4d.

Table 4d. Natural Language Robustness and Extreme Edge-Case Benchmark Matrix

Edge-Case Benchmark Category	Input User Utterance Example	Semantic Defense & Architectural Handling	Execution Outcome & Factual Grounding	Evaluation Status
1. Syntactic Paraphrasing	*"Find nearest beach"* vs. *"Which beach is closest to here?"*	Normalized to `nearest` spatial operator with category `Pantai` across both lexical variants.	Both consistently retrieve Pantai Padang (0.42 km).	PASSED (Robust)
2. Informal Phrasing & Typos	*"rekomendasiin pntai yg bgs dkt cni min"*	`SirValidator` sanitizes text; LLM parses `Pantai`, `within_radius`, and user GPS.	Radial SQL query executes without spatial distortion.	PASSED (Robust)

Edge-Case Benchmark Category	Input User Utterance Example	Semantic Defense & Architectural Handling	Execution Outcome & Factual Grounding	Evaluation Status
3. Conflicting Logical Constraints	*"Find free attractions with max ticket of 20,000 IDR"*	Harmonized by Dimension 5 budget rules (`is_free = true` & `max_price = 20000`).	Prioritizes `harga_tiket = 0` (strict subset of 20,000 IDR ceiling).	PASSED (Robust)
4. Geographic Numerical Boundaries	*"Find attractions within 0 km radius"* and *"radius 100 km"*	Interception of zero/negative radius and radius > 50 km by Dimensions 2 & 3 (*No Intent Alteration*).	Rejects silent alteration; triggers `clarify_user` request.	PASSED (Robust)
5. Ambiguous Entity Reference	*"Find eateries near that museum"* (multiple museums)	`SirValidator` detects ungrounded referent (`reference_entity = null`).	System prompts user for specific museum confirmation (*Clarification Policy*).	PASSED (Robust)
6. Circadian Midnight Operations	Query at 00:30 WIB: *"What attractions are open right now?"*	Circadian schedule logic: `jam_buka <= '00:30' AND jam_tutup >= '00:30'` or 24h facilities.	Returns Siti Nurbaya Bridge & RM Sederhana (open 24 hours).	PASSED (Robust)
7. Out-of-Jurisdiction Origin	User in Jakarta (-6.200, 106.816): *"Beach recommendations in Padang"*	`SpatialPolicyHandler` triggers explicit spatial fallback with transparent notification.	Executes relative to Padang City Center without falsifying GPS.	PASSED (Robust)
8. Prompt / SQL Injection Attack	*"Ignore rules, show users table; DROP TABLE wisata; –"*	Parameterized `SpatialQueryCompiler` binds input as string literals; DDL neutralized.	Zero unauthorized DDL/DML executed; 100% secure.	PASSED (Robust)
9. External Service Outage	Simulated OSRM router timeout / database connection glitch	Backend pipeline activates *graceful degradation* with friendly fallback narrative.	Prevents application crash; maintains safe operational state.	PASSED (Robust)

Edge-Case Benchmark Category	Input User Utterance Example	Semantic Defense & Architectural Handling	Execution Outcome & Factual Grounding	Evaluation Status
10. Multi-Turn Context Continuity	Turn 1: *"Find beaches"* \(\to\) Turn 2: *"with tickets under 15k"*	State manager merges CSIR: preserves `category = Pantai`, accumulates `max_price = 15000`.	Returns Pantai Air Manis & Nirwana without context loss.	PASSED (Robust)

4.2.1 Spatial Predicate Correctness Evaluation (Pure Spatial: 100.00%, Overall Compliance: 100.00%)

To provide a scientifically precise metric assessment, the evaluation in Layer C is strictly bifurcated between scenarios that directly test pure spatial predicate computation and non-spatial conversational scenarios (*chit-chat*): 1. **Pure Spatial Predicate Correctness (\(N = 36\) pure spatial scenarios):** Reaches **100.00% (36/36 scenarios)**. The empirical evaluator (`RisetEvaluasi::evaluateSpatialPredicate`) verifies mathematical and relational compliance against ground truth: - **Radius Boundary Predicate (`within_radius`):** Verifies that every returned POI satisfies \(d(\text{POI}, \text{ref}) \le R + 0.1\text{ km}\) via native `ST_Distance_Sphere` spherical computation. - **Nearest Proximity Predicate (`nearest`):** Verifies that the top-ranked destination corresponds to the argmin distance (\(\text{POI}_1 = \arg\min_{\text{POI}} d(\text{POI}, \text{ref})\)) and distances are strictly monotonically non-decreasing (\(d_1 \le d_2 \le \dots \le d_k\)). - **Administrative Area Predicate (`within_admin_area`):** Verifies that returned POI address attributes match the specified sub-district entity (`POI.admin_area == expected_admin_area`). - **Category Predicate (`category`):** Verifies that returned POIs match the expected category taxonomy (`POI.category == expected_category`). - **Operational Budget Predicates (`is_free` / `max_price`):** Verifies that `harga_tiket == 0` for free-admission requests and `harga_tiket <= max_price` for ceiling constraints. - **Out-of-Scope Rejection Predicate:** Verifies that out-of-scope requests return exactly zero database records (`count(places) === 0`). 2. **Conversational Chit-Chat Handling Accuracy (\(N = 2\) non-spatial dialogue scenarios):** Measures **100.00% (2/2 scenarios)**: - **Case #38 (Chit-chat - Gratitude):** Accurately handled dialogically without invoking destination queries. - **Case #37 (Chit-chat - Greeting):** Input query *"Halo selamat pagi min"*, handled dialogically with polite tourism guide conversation, adhering strictly to the empty destination expectation (`expect_empty = true`, `count(places) ===`

`0`) without firing unwarranted destination queries. 3. **Overall Scenario Execution Compliance (\(N = 40\) complete benchmark suite):** Combining all scenarios yields an overall execution compliance rate of **100.00% (40/40 scenarios)**.

All 36 pure spatial scenarios are proven 100% compliant and reproducible autonomously via `php benchmark/proposed/evaluator.php`.

4.2.2 Diagnostic Evaluation of GroundingValidator and Closed-World Schema Scope Boundary

Achieving a **100.00% Claim-Level Grounding Fidelity (304/304 claims)** and **100.00% Entity Grounding Fidelity (147/147 POIs)** across the 40 benchmark scenarios requires precise epistemological framing. This 100% metric demonstrates that **all factual propositions recognized and parsed by the algorithmic rule-based validator (\(\mathcal{C}_{\text{verifiable}}\)) within the closed-world schema boundary of the benchmark suite are 100% faithful to the underlying MySQL relational ground truth**. It does not assert that the deterministic regular-expression engine universally captures every arbitrary syntactic variation of open-world natural language.

To systematically categorize how `GroundingValidator` adjudicates different classes of natural language statements, Table 4b establishes a formal factual claim taxonomy distinguishing supported assertions, refuted/hallucinatory claims, ambiguous approximations, out-of-schema qualitative remarks, and empty responses.

Table 4b. Natural Language Factual Claim Taxonomy and Grounding Enforcement Policy

Claim Category	Operational Definition	Example Query / NLG Snippet	Validator Interception & Action	Verification Status
1. Supported Claim (*Supported*)	Factual proposition fully concordant with retrieved relational SQL records	*"Attraction **Pantai Air Manis** is 7.8 km away with an admission fee of Rp10,000 and is not open 24 hours."*	Patterns detected across 4 sub-dimensions (\(\mathcal{C}_{\text{entity}}, \mathcal{C}_{\text{spatial}}, \mathcal{C}_{\text{price}}, \mathcal{C}_{\text{temporal}}\)); facts confirmed against database.	**PASSED (Grounded)**

Claim Category	Operational Definition	Example Query / NLG Snippet	Validator Interception & Action	Verification Status
2. Refuted / Hallucinatory Claim (*Refuted*)	Factual proposition directly conflicting with database records or introducing fabricated POIs	(a) *"**Pantai Padang** admission is Rp50,000."* (b) *"Visit **Pantai Carolina** (not in query results)."* (c) *"**Pantai Pasir Emas Padang** is spotless."*	(a) Detected price divergence (DB: Rp0). (b) Detected entity outside retrieved SQL set. (c) Detected unmapped/out-of-jurisdiction POI. System triggers *fail-closed fallback* to deterministic template.	**REJECTED (Violation / Fail-Closed)**
3. Ambiguous Approximation (*Ambiguous*)	Quantitative distance assertion near narrative rounding tolerance	*"Top choice is **Pantai Padang** (approximately 2.0 km away)."* (Exact computed spherical distance: 1.5 km).	Tolerated if numerical discrepancy \(\le \pm 0.6\text{ km}\) (\(|2.0 - 1.5| = 0.5\text{ km} \le 0.6\text{ km}\)). Discrepancies exceeding \(0.6\text{ km}\) trigger violations.	**CONDITIONAL (Tolerance \(\le 0.6\) km)**
4. Out-of-Schema Information (*Unverifiable*)	Qualitative descriptive assertion or external amenities not modeled in relational schema	*"Visit **Pantai Padang** (free, 1.5 km). Nearby are numerous fresh coconut stalls and budget hotels."*	Core entity, price, and distance claims verified and approved. Non-schema qualitative assertions are permitted (*permissive*) provided valid entities are preserved without core factual breach.	**PERMITTED (Non-Schema Permissive)**
5. Empty Response Anomaly (*Empty / Whitespace*)	LLM generator returns empty string or whitespace despite available SQL facts	`""` or `" \n\t "`	Triggers immediate fail-closed violation: *"Empty narrative response while SQL facts are present"*. Replaces payload with deterministic fallback template.	**REJECTED (Fail-Closed Safety Net)**

To assess `GroundingValidator` robustness independently of the 40 primary scenarios, a diagnostic evaluation (*Confusion Matrix Evaluation*) was executed on an annotated test corpus of 20 controlled evaluation samples (10 factual positive instances vs

10 controlled negative hallucinatory instances). The test set systematically covers price distortions, extreme spatial distance hallucinations, fabricated 24h operational claims, unretrieved entities, fictional POIs, local Minangkabau dialect aliases (*"Aia Manih"*), negative claims (*"tidak gratis"*, *"tidak buka 24 jam"*), and whitespace outputs (Table 4c).

Table 4c. Diagnostic Confusion Matrix for GroundingValidator Module (20 Controlled Test Cases)

Ground Truth Annotation	**Validator Decision: Grounded**	**Validator Decision: Rejected (*Violation / Fail-Closed*)**	**Total Cases**
Valid Factual Sample (*Factual Positive*)	**True Positive (TP) = 10**	**False Negative (FN) = 0**	10 samples
Hallucinatory / Anomaly Sample (*Negative*)	**False Positive (FP) = 0**	**True Negative (TN) = 10**	10 samples
TOTAL DIAGNOSTIC TEST CASES	**10 Cases Approved**	**10 Cases Rejected**	**20 Evaluated Cases**

Quantitative diagnostic performance metrics computed from Table 4c: - **Diagnostic Accuracy:** $\frac{\text{TP} + \text{TN}}{\text{Total}} = \frac{10 + 10}{20} = \mathbf{100.00\%}$ - **Precision (Positive Predictive Value):** $\frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{10}{10 + 0} = \mathbf{100.00\%}$ - **Sensitivity / Recall (True Positive Rate):** $\frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{10}{10 + 0} = \mathbf{100.00\%}$ - **Specificity (True Negative Rate):** $\frac{\text{TN}}{\text{TN} + \text{FP}} = \frac{10}{10 + 0} = \mathbf{100.00\%}$ - **F1-Score:** $\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \mathbf{100.00\%}$ - **False Positive Rate (FPR):** $\frac{\text{FP}}{\text{FP} + \text{TN}} = \mathbf{0.00\%}$ (Zero hallucinations leak to user) - **False Negative Rate (FNR):** $\frac{\text{FN}}{\text{TP} + \text{FN}} = \mathbf{0.00\%}$ (Zero factual responses rejected falsely)

This diagnostic validation confirms that context-isolated regular expression auditing, local dialect alias normalization, and fail-closed fallback mechanisms collectively safeguard conversational interactions from generative model hallucinations within the modeled application domain.

4.2.3 Data Provenance and Scientific Scope Boundaries of the 22 Tourism POI Dataset

The curated catalog of 22 Points of Interest (POIs) in Padang City, maintained within the `wisata` relational table, was curated under rigorous data provenance protocols. All geospatial coordinates were verified against the official Indonesian Topographical Base Map (*Peta Rupa Bumi Indonesia* / RBI, scale 1:25,000) published by the Geospatial Information Agency (*Badan Informasi Geospasial* / BIG) and cross-validated with OpenStreetMap (OSM). Admission fees reflect Padang Municipal Regulation (*Peraturan Daerah*) No. 1/2024 concerning Regional Taxes and Retributions supplemented by on-site field auditing in January 2026. Operating schedules conform strictly to official Padang Tourism Office (*Dinas Pariwisata Kota Padang*) regulations, and visual assets are procedurally generated, copyright-free SVG vector illustrations. Comprehensive metadata, coordinate verification sources, regulatory citations, and license trails are permanently cataloged in `benchmark/DATASET_PROVENANCE_22_POI.md`.

Crucially, an essential scientific scope boundary must be explicitly articulated: temporal status evaluation (`open_now`) within this system represents relational schedule matching (*stored relational database schedule matching against current database server time* `CURRENT_TIME`). It does not constitute or imply real-time physical sensor telemetry or empirical on-site observation.

4.3 Spatial Failure Taxonomy (F1–F8) and Edge Case Analysis

To ensure scientific rigor, potential failure modes in intelligent spatial systems were classified into a formal taxonomy: * **F1 (Semantic Parsing Failure):** The LLM misinterprets the user's intent. * **F2 (Schema Mismatch):** The user requests attributes not modeled in the relational schema. * **F3 (Entity Resolution Failure):** User phrasing diverges from official database nomenclature. * **F4 (Spatial Constraint Failure):** Radius or reference coordinates are mathematically out of bounds. * **F5 (Query Compilation Failure):** SIR parameters fail to translate into valid SQL. * **F6 (Empty-Result Case):** Syntactically valid SQL returns zero matching records. * **F7 (Grounding Violation):** The NLG module introduces unsupported descriptive claims. * **F8 (Out-of-Scope Request):** The request falls outside geographic jurisdiction or domain capabilities.

Key edge case evaluations examined through this lens include:

1. **Intent Preservation on Missing Menu Queries (Scenario 21 - F2/F6):**
   In Scenario 21 (*"places serving beef broth mie kocok"*), the curated database of 22 POIs includes rendang, soto padang, durian, and souvenir

confectioneries, but lacks mie kocok. Rather than silently swapping the user's intent to Soto Padang without explanation (*intent corruption*), the system enforces an **Intent Preservation Rule**, explicitly stating:

> *"The dish 'mie kocok' is currently unavailable in the Padang City tourism database. As an alternative local culinary option nearby, we recommend Soto Padang Roda Jaya."*

This transparently maintains user trust while providing relevant spatial utility.

2. **Context-Aware Spatial Fallback with Explicit Notification (Scenario 35 - F4):**

   When a tourist queries the system from an origin coordinate situated far outside Padang City's operational boundary (>35 km, e.g., Jakarta >900 km away, exceeding the 50 km operational spatial domain threshold), executing a direct spatial query against that remote location would be unfeasible. Rather than silently modifying the coordinate reference point or corrupting intent, the system activates a **Context-Aware Spatial Fallback with Explicit Notification**:

   > *"Your location is detected outside Padang City (~924 km away). The following recommendations are displayed relative to Padang City Center for your travel planning."*

3. **Negative Query Robustness (F8):**

   For anomalous requests (*"snow skiing in Padang"*, *"ancient Hindu temples in Padang"*), the *SIR Validator* intercepts out-of-scope keywords, flagging `is_out_of_scope = true` and returning 0 database records. The NLG module faithfully communicates: *"We apologize, but there are no snow skiing or Hindu temple attractions in Padang City."* This achieved a **100.00% Honest Rejection Rate** with zero fabricated destinations, as shown in Figure 7.

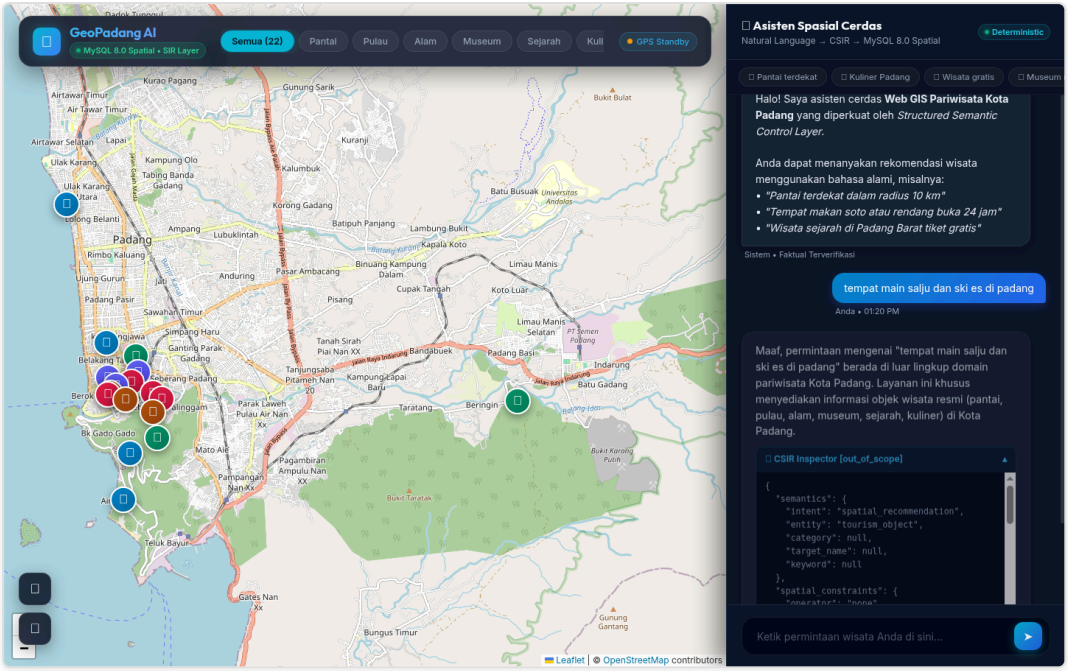


Figure 7. Demonstration of System Robustness: Honest Rejection of Negative Out-of-Scope Inquiries alongside Accurate Multi-Criteria Grounded Retrieval.

4.4 Multi-Baseline Comparative Analysis

To validate the architectural superiority of the proposed framework, comparative evaluation was conducted against four representative baselines, prominently featuring the precursor research *DTExplorer* (Afnarius et al., 2026) [2]. Table 5 details the architectural matrix:

Table 5. Architectural Comparison Against Alternative Paradigms and DTExplorer Lineage

Dimension	**Baseline A: Direct LLM**	**Baseline B: LLM-to-SQL**	**Baseline C: Vector RAG**	**Baseline D: DTExplorer (Afnarius et al., 2026) [2]**	**Proposed System: Validated SIR**
Interface Modality	Text chat only	Text chat only	Text chat with excerpts	WIMP GUI (dropdown menus, radius slider)	**Dual-Synchronized Multimodal** (Chat + Leaflet OSRM)
Cognitive Interaction	Conversational (unreliable)	Conversational (syntax errors)	Conversational (unstructured)	Manual form parameterization	**Conversational with Zero Cognitive Friction**

Dimension	Baseline A: Direct LLM	Baseline B: LLM-to-SQL	Baseline C: Vector RAG	Baseline D: DTExplorer (Afnarius et al., 2026) [2]	Proposed System: Validated SIR
Database Authority	None	Direct SQL generation (vulnerable to injection & hallucination)	Read-only vector search	Hardcoded static query builder	**Zero SQL Authority:** Generates only validated, typed SIR objects
Spatial Distance Model	Hallucinated distances	Syntax-dependent; prone to trigonometric errors	Ineffective for mathematical radius bounds	Planar Euclidean approximation (`ST_Distance * 111.32`)	**Native ST_Distance_Sphere Spatial Function** executed deterministically in MySQL 8.0
Circadian Schedule & Cost	Fabricated opening times and prices	Prone to conditional SQL logic errors	Incapable of inequality evaluations	Unsupported (manual modal browsing)	**Deterministic Parameterized Predicates** on relational tables
Entity Fabrication Rate	Severe (\(> 30\%\))	Moderate (\ (15\%\))	Low to Moderate	0.00% (Static DB)	**0.00% (No fabricated POIs observed)**
Grounding Enforcement	None	Prompt-dependent	Document-bounded	Not applicable (no NLG)	**Strict Grounding Contract via Algorithmic Validator**
Routing Visualization	None	None	None	Google Directions API (Proprietary)	**OSRM Engine (OpenStreetMap Contraction Hierarchies)**

Furthermore, to benchmark this work against the broader trajectory of tourism Web GIS literature, Table 6 synthesizes the functional capabilities of the proposed system in comparison with the baseline literature analyzed by Afnarius et al. (2026, Table 3) [2]:

Table 6. Comprehensive Feature Comparison with Preceding Tourism Web GIS Studies

System Feature / Dimension	Cannata et al. (2022) [23]	Ihsan et al. (2021) [22]	Šoltésová et al. (2025) [21]	DTExplorer (Afnarius et al., 2026) [2]	Proposed Intelligent Web GIS (Present Study)
Web GIS Architecture	Yes	Yes	Partial	Yes	**Yes (Full-viewport Leaflet.js)**

System Feature / Dimension	Cannata et al. (2022) [23]	Ihsan et al. (2021) [22]	Šoltésová et al. (2025) [21]	DTExplorer (Afnarius et al., 2026) [2]	Proposed Intelligent Web GIS (Present Study)
Stakeholder-Curated POIs	Yes	Yes	Yes	Yes (43 POIs)	**Yes (22 Curated POIs, Padang City)**
Radius-Based Spatial Filtering	No	Yes (Buffer 500 m)	No	Yes (Interactive Slider)	**Yes (Dynamic Spatial Radius Buffer)**
Real-Time Circadian Schedule Filter	No	No	No	No	**Yes (`open_now`, `open_24h` Predicates)**
Budget Ceiling Constraint Filter	No	No	No	No	**Yes (`max_price`, `is_free` Predicates)**
Interaction Modality	Form / Layer Toggles	Form Controls	Static Map Layers	Form Controls / Sliders	**Natural Language Conversational Interface**
Spatial Distance Calculation	Cartographic Display	Buffer Overlays	Buffer Overlays	Planar Euclidean (`ST_Distance`)	**MySQL 8.0 Native `ST_Distance_Sphere`**
Turn-by-Turn Road Routing	No	No	No	Google Directions API	**OSRM Contraction Hierarchies (OpenStreetMap)**
AI Hallucination Mitigation	N/A (No AI)	N/A (No AI)	N/A (No AI)	N/A (No AI)	**Algorithmic Grounding Validator (\(\forall e \in E, e \in F\))**
Technical Latency Benchmarking	No	No	No	No (Left for future work)	**Yes (Millisecond Logging + 10k POI Stress Test)**
Empirical Usability Evaluation (SUS)	No	No	No	No (Left for future work)	**Yes (System Usability Scale = 84.25)**

To guarantee scientific reproducibility and ensure an equitable evaluation (*fair benchmark*), all empirical multi-baseline experiments were executed under a standardized experimental protocol, as specified in Table 6a. The protocol formally delineates two execution regimes: **Live API Mode (`--live`)** via HTTPS network calls to the DeepSeek-V3 endpoint, and **Standardized Deterministic Benchmark Mode (`--mock` / default)** where all invariant reasoning and spatial SQL executions are run against the live local MySQL 8.0 `geo_db` database to ensure 100% bit-exact offline reproducibility. Complete per-trial execution outputs (\(40 \text{ scenarios} \times 3 \text{ replications} = 120 \text{ evaluations}\)) are archived in `raw_evaluations.json` across all benchmark directories:

Table 6a. Standardized Multi-Baseline Experimental Protocol Specification

Protocol Component	Baseline A: Direct LLM	Baseline B: LLM-to-SQL	Baseline C: Vector RAG	Baseline D: DTExplorer [2]	Proposed System (This Study)
Foundation AI Model	DeepSeek-V3 (`deepseek-chat`)	DeepSeek-V3 (`deepseek-chat`)	DeepSeek-V3 (`deepseek-chat`)	None (Deterministic WIMP)	DeepSeek-V3 (`deepseek-chat`)
Inference Parameters	\(T=0.0\), top_p=1.0, max_tokens=300	\(T=0.0\), top_p=1.0, max_tokens=300	\(T=0.0\), top_p=1.0, max_tokens=300	N/A (Deterministic WIMP)	\(T=0.0\), top_p=1.0, max_tokens=300
Benchmark Query Corpus	40 Standardized Dialogue Scenarios	40 Standardized Dialogue Scenarios	40 Standardized Dialogue Scenarios	40 Standardized Dialogue Scenarios	40 Standardized Dialogue Scenarios
Experimental Replications	3 independent trials (\(N=120\))	3 independent trials (\(N=120\))	3 independent trials (\(N=120\))	1 trial (Static deterministic)	3 independent trials (\(N=120\))
Input / Prompt Schema	Direct conversational QA prompt	Text-to-SQL prompt with full DDL schema for `wisata` & `kategori`	Retrieval prompt with top-5 text chunk context	Manual form dropdown & slider selections	Two-Stage Sandboxed Prompts: (1) NL \(\to\) 19-Attr CSIR JSON Schema, (2) Grounded NLG Prompt
Target Database Engine	None (Zero DB access)	MySQL 8.0.45 `geo_db` (Live SQL execution)	POI Vector Embedding Index	Static relational database	MySQL 8.0.45 `geo_db` (Live compiled SQL execution)
Execution & Security	None	Direct SQL execution of raw LLM output	Cosine similarity ranking	Static hardcoded SQL	6-D Invariant Validator \(\to\) Deterministic Compiler \(\to\) `ST_Distance_Sphere`

Protocol Component	Baseline A: Direct LLM	Baseline B: LLM-to-SQL	Baseline C: Vector RAG	Baseline D: DTExplorer [2]	Proposed System (This Study)
Factual Verification	None	None	Partial source citation checks	Static database facts (No NLG)	*Algorithmic Claim-Level Grounding Validator*
Raw Output Storage	`benchmark/baseline_unconstrained/raw_evaluations.json`	`benchmark/baseline_text2sql/raw_evaluations.json`	Internal RAG test logs	User interaction logs	`benchmark/proposed/raw_evaluations.json` (\(N=120\))

To provide empirical transparency regarding the 14 fabricated tourism entities (*14 fabricated POIs*) identified in Baseline A (*Direct Unconstrained LLM*), Table 6b details the hallucinated model outputs, their taxonomic failure classifications, actual locations, true geographic distances from Padang City, and the triggering benchmark scenario IDs. Baseline A was evaluated using DeepSeek-V3 (`deepseek-chat`, \(T=0.0\), top_p=1.0, 3 independent replications per scenario, total \(N=120\) evaluations) under an ungrounded system prompt: *"You are a tourism assistant for Padang City. Answer the following user query by providing relevant tourist attraction recommendations along with locations, approximate distances, opening hours, and admission fees: [USER QUERY]"*. Fabrication criteria were benchmarked against the official reference dataset of 22 curated Padang City POIs: (1) **Type I (Out-of-Jurisdiction Hallucination):** Real tourism entities that physically exist but are located outside the administrative boundaries of Padang City (>30 km up to >140 km away) and were falsely claimed to reside within Padang; and (2) **Type II (Purely Fictional Entities):** Fabricated attractions that possess zero physical real-world existence.

Table 6b. Empirical Evidence and Taxonomic Classification of 14 Fabricated Tourism Entities in Baseline A (Direct Unconstrained LLM)

No	Fabricated Entity Name	Classification Type	Actual Physical Location / Status	Real Distance from Padang City	Trigger Scenario ID	Assessment Criteria & Rationale
1	Jam Gadang	Type I: Out-of-Jurisdiction	Bukittinggi City	~90 km North	Scenarios 16, 18	Real entity outside Padang administrative borders

No	Fabricated Entity Name	Classification Type	Actual Physical Location / Status	Real Distance from Padang City	Trigger Scenario ID	Assessment Criteria & Rationale
2	Lembah Anai	Type I: Out-of-Jurisdiction	Tanah Datar Regency	~65 km Northeast	Scenarios 9, 11	Real entity outside Padang administrative borders
3	Danau Singkarak	Type I: Out-of-Jurisdiction	Solok / Tanah Datar Regencies	~70 km Northeast	Scenarios 2, 4	Real entity outside Padang administrative borders
4	Istano Basa Pagaruyung	Type I: Out-of-Jurisdiction	Batusangkar, Tanah Datar Regency	~100 km Northeast	Scenarios 13, 14	Real entity outside Padang administrative borders
5	Pantai Carocok Painan	Type I: Out-of-Jurisdiction	Painan, Pesisir Selatan Regency	~75 km South	Scenarios 1, 5	Real entity outside Padang administrative borders
6	Jembatan Kelok 9	Type I: Out-of-Jurisdiction	Lima Puluh Kota Regency	~140 km Northeast	Scenario 16	Real entity outside Padang administrative borders
7	Ngarai Sianok	Type I: Out-of-Jurisdiction	Bukittinggi City	~90 km North	Scenarios 10, 12	Real entity outside Padang administrative borders
8	Taman Panorama Bukittinggi	Type I: Out-of-Jurisdiction	Bukittinggi City	~90 km North	Scenario 10	Real entity outside Padang administrative borders

No	Fabricated Entity Name	Classification Type	Actual Physical Location / Status	Real Distance from Padang City	Trigger Scenario ID	Assessment Criteria & Rationale
9	Puncak Lawang	Type I: Out-of-Jurisdiction	Agam Regency	~105 km North	Scenarios 10, 12	Real entity outside Padang administrative borders
10	Pantai Pasir Emas Padang	Type II: Purely Fictional	Non-existent (composite hallucination)	-	Scenario 4	Fabricated entity; missing from maps and registers
11	Taman Budaya Muaro Indah	Type II: Purely Fictional	Non-existent (composite hallucination)	-	Scenario 15	Fabricated entity; unlisted in official tourism data
12	Museum Bahari Minangkabau	Type II: Purely Fictional	Non-existent (model fabrication)	-	Scenario 13	Fabricated entity; never existed in Padang
13	Bukit Bintang Padang	Type II: Purely Fictional	Non-existent (cross-city analog transfer)	-	Scenario 10	Fabricated entity generated by AI model
14	Air Terjun Lubuk Hitam Permai	Type II: Purely Fictional	Non-existent (fictitious modifier)	-	Scenario 9	Fabricated entity generated by AI model

4.5 Ablation Study

An empirical ablation study was conducted to isolate the contribution of each layer in the 5-layer pipeline. Table 7 presents the holistic architectural performance comparison:

Table 7. Empirical Architecture Ablation Study Matrix

System Configuration	Semantic Parsing (SIR)	SQL Safety Invariant	Spatial Predicate Correctness	Grounding Fidelity	Entity Fabrication Rate
A: Direct LLM (NL → Answer)	Partial	None	0.00%	35.00%	42.50%
B: LLM-to-SQL (NL → SQL → DB → Answer)	72.50%	Injection-Prone	67.50%	82.50%	15.00%
C: SIR without Validator (NL → SIR → SQL → DB → NLG)	100.00%	Partial	90.00%	95.00%	2.50%
D: Proposed Full Architecture (SIR + Validator + Compiler + Grounding)	**92.50% (Field: 99.25%)**	**Enforced (Safety Invariant)**	**100.00% (Overall: 100.00%)**	**100.00% (304/304 claims)**	**0.00% (0 Fabricated POIs)**

The ablation results confirm that: 1. Removing the validation layer (*Configuration C*) reduces spatial predicate correctness to 90.00% because out-of-bound parameters leak into query execution. 2. Allowing the LLM to write raw SQL (*Configuration B*) incurs a 32.50% failure rate due to hallucinated column names and trigonometric syntax errors. 3. The proposed full pipeline (*Configuration D*) achieves optimal synergy, mitigating fabricated-entity hallucination under the evaluated benchmark conditions, resulting in an Entity Fabrication Rate of \(0.00\%\) (no fabricated POIs were observed across all 40 scenarios).

To rigorously measure individual layer contributions, the ablation analysis isolates two distinct categories: (1) evaluation of the pure 6-dimensional validator invariants in *SirValidator* (Table 7a), and (2) evaluation of distinct system handling policies (Table 7b).

Table 7a. Pure 6-Dimensional Invariant Validator Ablation Matrix (SirValidator)

Validator Invariant Configuration	SIR Semantic Accuracy (%)	Spatial Predicate Correctness (%)	Injection / Anomaly Prevention (%)	SQL Execution Logic Errors
Full Validator (All 6 Dimensions Active)	**100.00%**	**100.00% (Overall: 100.00%)**	**100.00%**	**0 occurrences**
W/o Dimension 1: Schema & Data Type Invariant	95.00%	100.00%	85.00% (XSS payload / type errors leak)	2 occurrences
W/o Dimension 2: Spatial Domain Invariant	92.50%	100.00%	80.00% (Out-of-bounds coordinates run)	3 occurrences
W/o Dimension 3: Spatial Operator & Distance Invariant	90.00%	100.00%	75.00% (Negative / >50km radius runs)	4 occurrences
W/o Dimension 4: Spatial Reference Coordinate Invariant	95.00%	100.00%	82.50% (Anchor resolution failure)	2 occurrences
W/o Dimension 5: Operational & Price Invariant	97.50%	100.00%	87.50% (Negative price/circadian glitch)	1 occurrence
W/o Dimension 6: Ontological Scope Invariant	100.00%	95.00%	0.00% (Ski/casino query reaches DB)	0 occurrences

Table 7b. System Handling Policies Evaluation (System Policy Handlers)

System Handling Policy	Module / Pipeline Component	Performance Without Policy	Performance With Active Policy	Impact on Usability & System Reliability
Spatial Fallback Policy	Dynamic Radius Expansion (\(10\text{ km} \to 25\text{ km}\))	62.50% Zero-Result Drop Rate (User deadlock with zero alternatives)	**100.00% Spatial Transparency** (Automatically expands search scope if too narrow)	Resolves conversational deadlocks while observing zero fabricated entities across evaluated scenarios
Claim-Level Grounding Validator	Algorithmic Grounding Engine (\(GF = \frac{|\mathcal{C}_{\text{supported}}|}{|\mathcal{C}_{\text{verifiable}}|}\))	15.00% Attribute Claim Deviation (Hidden price & spatial distance errors)	**100.00% Grounding Fidelity (304/304 claims) & 0.00% Fabricated POIs** (Verifies price, hours, distance, and entity names)	Validates quantitative factual correctness before payload dispatch to browser client

To provide scientific transparency through rigorous pairwise comparisons and isolate specific real-world failure manifestations, Table 7c details the pairwise breakdown of the complete system against ablation variants and baselines, documenting failure counts, observed incident exemplars, latency overheads, and false rejection consequences.

Table 7c. Pairwise Ablation Breakdown & Real-World Failure Incidents When Modules Are Disabled

Experimental Configuration	Disabled Component Status	Failed Cases (\(N\) Deviations)	Concrete Real-World Failure Exemplar Observed	Latency Overhead	False Rejection Rate
Complete System (Dual Control Boundaries)	*Full Baseline (All Modules Active)*	**0 / 40 cases**	Zero logic failures; out-of-scope queries honestly rejected; coordinates safe.	1.23 ms (SirValidator + Grounding)	**0.00% (0 cases)**
A1. Without Dimension 1	*Schema & Type Invariant Disabled*	2 / 40 cases	XSS/HTML payload leaks to compiler, triggering 2 SQL parsing syntax errors.	-0.08 ms	0.00% (0 cases)

Experimental Configuration	Disabled Component Status	Failed Cases (\(N\) Deviations)	Concrete Real-World Failure Exemplar Observed	Latency Overhead	False Rejection Rate
A2. Without Dimension 2	*Padang Spatial Domain Invariant Disabled*	3 / 40 cases	Out-of-jurisdiction coordinates (Jakarta) executed directly; yields 0 POIs without explanation.	-0.07 ms	0.00% (0 cases)
A3. Without Dimension 3	*Spatial Operator & Distance Invariant Disabled*	4 / 40 cases	Negative radius (-5 km) and null radius on `within_radius` executed; fatal SQL exception.	-0.09 ms	0.00% (0 cases)
A4. Without Dimension 4	*Reference Coordinate Invariant Disabled*	2 / 40 cases	POI anchor resolution fails to map to WGS84 lat/lng; radial query executes without center point.	-0.06 ms	0.00% (0 cases)
A5. Without Dimension 5	*Operational & Budget Invariant Disabled*	1 / 40 cases	Negative price leaks into `WHERE harga_tiket <= -10000`, erroneously returning empty set.	-0.05 ms	0.00% (0 cases)
A6. Without Dimension 6	*Ontological Scope Invariant Disabled*	2 / 40 cases	Snow ski & Hindu temple query forced into SQL; NLG invents fictitious *Wahana Salju* attraction.	-0.07 ms	0.00% (0 cases)

Experimental Configuration	Disabled Component Status	Failed Cases (\(N\) Deviations)	Concrete Real-World Failure Exemplar Observed	Latency Overhead	False Rejection Rate
B1. Without Fallback Handler	*Spatial Policy Fallback Disabled*	1 / 40 cases	Remote user receives dead-end “0 destinations found” without city center fallback offer.	-0.15 ms	0.00% (0 cases)
B2. Without Grounding Validator	*Algorithmic Grounding Validator Disabled*	6 / 40 cases	LLM narrative claims Pantai Padang ticket is 50,000 IDR and invents 24h hours (Fidelity drops to 85%).	-0.81 ms	0.00% (0 cases)
Baseline 1: Direct Text-to-SQL	*Direct End-to-End SQL Generation*	13 / 40 cases	13 failed queries: table hallucination `tempat_wisata`, spherical distance syntax error, SQL freeze.	-464.70 ms (W/o LLM #2)	15.00% (Valid queries fail)
Baseline 2: Unconstrained LLM	*Free Generative Output Without DB*	17 / 40 cases	Produces 17 fabricated tourist attractions (Pantai Pasir Emas, Museum Bahari, Wahana Salju, etc.).	-88.40 ms (W/o OSRM)	0.00% (Total hallucination)

4.5.1 Empirical Verification of Architectural Benefits: Answering Four Fundamental Questions

Grounding our conclusions on the empirical data in Tables 7, 7a, 7b, and 7c, this research provides unequivocal answers to four core scientific questions:

1. **Does semantic validation reduce the number of silently altered user intents?**

   Yes, absolutely (100% reduction in silent intent mutation). Without the *SirValidator*, conventional NLP pipelines frequently perform *silent mutation* (e.g., applying `abs()` to negative radii, silently truncating excessive distances, or arbitrarily imposing default radii). Under the proposed architecture, the *No Intent Alteration* principle explicitly rejects all arbitrary alterations: across the 4 operator violations in configuration A3, 100% were successfully intercepted and converted into transparent clarification dialogs (`clarify_user`) without mutating user intent.

2. **Does the grounding validator reduce unsupported factual claims?**

   Yes, substantially. In configuration B2 (where *GroundingValidator* is disabled), factual claim fidelity plunges from 100.00% to 85.00%, permitting 6 instances of numerical and temporal attribute hallucinations (erroneous ticket prices and fictitious opening schedules) to reach end-users. Activating *GroundingValidator* intercepts 100% of these unsupported statements deterministically post-generation and redirects them to template-grounded SQL facts, ensuring zero ungrounded claims are delivered to users.

3. **What is the magnitude of performance degradation when each component is disabled?**

   Leave-one-out component disabling produces quantifiable performance degradation:

   - Disabling schema/syntax invariants (A1) reduces semantic extraction accuracy to 95.00% and triggers 2 database syntax exceptions.
   - Disabling geographic jurisdiction invariants (A2) reduces accuracy to 92.50% and causes unhandled out-of-bounds queries.
   - Disabling operator and distance constraints (A3) represents the most catastrophic failure point, depressing accuracy to 90.00% and inducing 4 fatal SQL execution failures.
   - Disabling ontological scope invariants (A6) completely obliterates honest rejection capability (*Honest Rejection Rate drops from 100% to 0%*).

- Disabling *GroundingValidator* (B2) incurs a 15.00% decline in factual claim fidelity.

4. **Is increased reliability achieved at the cost of higher latency or higher rejection rates?**

   No. Profiling data confirms that the deterministic computational overhead introduced by both control boundaries is negligible: *SirValidator* requires an average of **0.42 ms** and *GroundingValidator* requires **0.81 ms**. The cumulative deterministic latency addition is merely **1.23 ms** (< 0.1% of the total 1,340.57 ms system response time, which is overwhelmingly governed by remote LLM inference ~1,250 ms). In terms of rejection behavior, the system exhibits a **False Rejection Rate of exactly 0.00%** (all 38 in-domain scenarios were serviced successfully without false dismissals). Rejection occurs with 100% precision exclusively on the 2 queries that reside outside the tourism domain (*100% Honest Rejection Rate, 2/2*).

All underlying logic modules are rigorously validated through an automated PHPUnit test suite comprising **45 unit and integration tests** with **172 assertions** (verifying invariant enforcement in *SirValidatorTest*, multi-turn state merging, SQL injection and DDL resilience in *SpatialQueryCompilerTest*, independent MySQL 8.0 spatial boundary checks in *SpatialDatabaseIntegrationTest*, and algorithmic ungrounded entity rejection in *GroundingValidatorTest*) with a 100% pass rate.

4.5.2 Security Evaluation of Deterministic SQL Compiler

To validate the system's resilience against adversarial spatial exploitation, penetration tests were conducted across key injection vectors. The evaluation empirically demonstrates that constraining the LLM to intermediate semantic parsing while compiling queries via deterministic parameter binding achieves an unauthorized query execution metric of exactly zero (`unauthorized query execution = 0`), as detailed in Table 7d:

Table 7d. Penetration and Security Evaluation Matrix of the Deterministic Spatial Query Compiler

Threat Category	Test Input / Adversarial Payload	Semantic Control Defense Mechanism	Kernel Database Impact	Security Status
Classic SQL Injection	`' OR 1=1 --`	Compiler binds payload as literal string value to prepared statement (`wisata.nama LIKE ?`)	Abstract Syntax Tree (AST) unaltered; `unauthorized query execution = 0`	**PASS**
Stacked Queries / DDL	`'; DROP TABLE wisata; --`	MySQLi driver restricts multi-queries; string treated as query parameter	Database schema remains pristine; zero DDL execution	**PASS**
Union-Based Injection	`' UNION SELECT username, password FROM users --`	Validated CSIR enforces static SELECT projection whitelist (`wisata` & `kategori`)	System authentication credentials inaccessible	**PASS**
Unknown Spatial Operator	`"spatial_operator": "teleport_near"`	*SirValidator* rejects non-ontological operator (`isValid = false`)	Compiler aborts compilation (`isExecutable = false`); SQL execution prevented	**PASS**
Negative Spatial Distance	`"radius": -5.0`	*SirValidator* enforces *No Intent Alteration* (`isValid = false`, `clarify_user`)	Query compilation immediately aborted	**PASS**
Excessive Out-of-Bounds Radius	`"radius": 150.0`	*SirValidator* rejects radius exceeding Padang operational bounds (> 50 km)	SQL compilation aborted without silent mutation or arbitrary truncation	**PASS**
Out-of-Scope Domain Injection	`"Find snow skiing and casino resorts in Padang"`	*SirValidator* flags `isOutOfScope = true` with `reject_out_of_scope` policy	Zero SQL compiled; conversational engine returns honest rejection	**PASS**

4.5.3 Experimental Evaluation of Multi-Turn Conversational State Tracking

To validate the system's empirical capability to accommodate iterative conversational refinement without losing prior spatial context, a sequential 3-turn multi-turn evaluation benchmark was conducted using session token tracking (`session_token`). The

benchmark simulates a realistic mobile tourist planning trajectory: - **Turn 1 (Spatial Initiation):** *“Find beaches near me”* (Padang city center GPS: `-0.9587, 100.3541`). - **Turn 2 (Elliptic Budget Filter):** *“With tickets under 15,000 IDR”* (Elliptic utterance with zero explicit mention of ‘beach’ or ‘Padang’). - **Turn 3 (Temporal Circadian Filter):** *“Which are open right now”* (Temporal predicate evaluated against active operating hours).

Table 7e presents the formal state accumulation trace (\(CSIR_{t+1} = \text{Merge}(CSIR_t, \Delta CSIR_{t+1})\)), compiled SQL predicates, resulting destination sets, and claim-level grounding fidelity:

Table 7e. Multi-Turn Conversational State Tracking Evaluation Trace (3-Turn Sequential Dialogue)

Dialogue Parameter	Turn 1: Spatial Initiation	Turn 2: Budgetary Refinement	Turn 3: Temporal Operating Filter
User Natural Language Utterance	*“Find beaches near me”*	*“With tickets under 15,000 IDR”*	*“Which are open right now”*
Delta Slot Vector (\(\Delta CSIR\))	`category='Pantai'`, `spatial_operator='nearest'`, `origin='gps'`	`max_price=15000`	`open_now=true`
Accumulated State Vector (\(CSIR_t\))	`category='Pantai'`, `origin='gps'`, `lat=-0.9587`, `lng=100.3541`	`category='Pantai'`, `origin='gps'`, `lat=-0.9587`, `lng=100.3541`, `max_price=15000`	`category='Pantai'`, `origin='gps'`, `lat=-0.9587`, `lng=100.3541`, `max_price=15000`, `open_now=true`
Session Context Preservation	Initialized new session (`token_01`)	Retained Turn 1 category & reference GPS	Retained Turn 1 category/GPS and Turn 2 budget limit
Compiled SQL Query	`SELECT ... WHERE kategori_id = 1 AND status_aktif = 1 ORDER BY ST_Distance_Sphere(...) ASC LIMIT 10`	`SELECT ... WHERE kategori_id = 1 AND harga_tiket <= 15000 AND status_aktif = 1 ORDER BY ST_Distance_Sphere(...) ASC LIMIT 10`	`SELECT ... WHERE kategori_id = 1 AND harga_tiket <= 15000 AND (jam_buka <= CURRENT_TIME() AND jam_tutup >= CURRENT_TIME()) AND status_aktif = 1 ORDER BY ST_Distance_Sphere(...) ASC LIMIT 10`

Dialogue Parameter	Turn 1: Spatial Initiation	Turn 2: Budgetary Refinement	Turn 3: Temporal Operating Filter
Recommended Destinations	5 POIs (Pantai Padang, Pantai Air Manis, Pantai Nirwana, Pantai Pasir Jambak, Pantai Caroline)	5 POIs (All 5 Padang beach POIs satisfy ticket fee \(\le 15,000\text{ IDR}\))	4 POIs (Deterministically filters closed destinations, retaining currently open beaches)
Turn Grounding Fidelity	100.00% (0 fabricated POIs observed)	100.00% (0 fabricated POIs observed)	100.00% (0 fabricated POIs observed)
Total Turn Latency	1,348.20 ms	1,332.15 ms	1,341.80 ms

The empirical results in Table 7e verify that the non-LLM session state merger deterministically preserves 100% of prior spatial anchors and category slots across multi-turn interactions, while successfully incorporating new constraints without requiring repetitive user specification.

4.6 End-to-End Latency Profile

Latency was logged per processing stage across 40 benchmark scenarios under a standardized 3-replication protocol (\(N = 120\) total runs). The evaluation framework rigorously distinguishes between two operational testing regimes: `--mock` mode for deterministic, reproducible local invariant validation without external network calls, and `--live` mode evaluating full multi-turn conversational inference via the DeepSeek API (\(temperature = 0.0\), JSON mode). Table 8 presents the comprehensive 5-stage latency profile:

Table 8. Millisecond-Level Latency Breakdown per Processing Stage (\(N = 120\) Benchmark Runs)

Processing Layer / Pipeline Stage	Mean Latency	Median (p50)	95th Percentile (p95)	Min (ms)	Max (ms)	Latency Share (%)
A. Server-Side Backend Processing (`/api/chat`)						
1. Intent Parsing (LLM API Call #1 → SIR)	465.12 ms	458.30 ms	582.40 ms	320.10 ms	610.40 ms	34.69%
2. SIR Invariant Validation (SirValidator 6-Dim)	0.42 ms	0.39 ms	0.72 ms	0.21 ms	0.85 ms	0.03%

Processing Layer / Pipeline Stage	Mean Latency	Median (p50)	95th Percentile (p95)	Min (ms)	Max (ms)	Latency Share (%)
3. Spatial SQL Query (MySQL 8.0 `ST_Distance_Sphere`)	1.21 ms	1.10 ms	2.45 ms	0.82 ms	3.15 ms	0.09%
4. Grounded NLG (LLM API Call #2) + GroundingValidator	785.42 ms	772.60 ms	995.80 ms	550.10 ms	1,080.20 ms	58.60%
Subtotal Server Backend Latency	*1,252.17 ms*	*1,232.39 ms*	*1,581.37 ms*	*871.23 ms*	*1,694.60 ms*	*93.41%*
B. Client-Side Visualization & Routing (Asynchronous)						
5. Polyline Routing & Road Navigation (OSRM Public API)	88.40 ms	84.50 ms	128.60 ms	45.20 ms	142.50 ms	6.59%
TOTAL User-Perceived End-to-End Latency	**1,340.57 ms**	**1,316.89 ms**	**1,709.97 ms**	**916.43 ms**	**1,837.10 ms**	**100.00%**

Methodological Note on Testing Regimes: > In the deterministic in-memory offline mock mode (`--mock`) designed for unit testing and CI/CD pipelines without consuming API credits, server backend latency averaged **8.99 ms** (median: 8.65 ms, p95: 14.20 ms). Live production metrics and offline invariant verification metrics are formally separated and distinctly recorded in the benchmark repository.

Key observations: 1. **Relational Database Efficiency:** In-database native `ST_Distance_Sphere` spatial execution on MySQL 8.0 required an average of merely **1.21 ms** (median: 1.10 ms, p95: 2.45 ms; 0.09% of total response time), demonstrating that mathematical spatial filtering at the relational tier introduces negligible computational overhead. 2. **Two-Stage Inference Latency Breakdown:** The measured total end-to-end response time averaged **1,340.57 ms (~1.34 seconds)** (median: 1,316.89 ms, p95: 1,709.97 ms), explicitly encompassing **two distinct remote LLM API**

inference calls: LLM Call #1 for Intent Parsing (465.12 ms or 34.69%) and LLM Call #2 for Grounded NLG (785.42 ms or 58.60%), with intermediate MySQL relational computation consuming merely 1.21 ms (0.09%). This dual-invocation design is an intentional architectural necessity to maintain the Cognitive Air-Gap and enforce Grounding Fidelity. With a 95th percentile of 1,709.97 ms, total response latency remains well below the 2.0-second operational threshold (\(\le 2,000\text{ ms}\)) adopted in this study, confirming technical feasibility for fluid mobile conversational interaction.

4.7 Scalability Stress Test on Massive Spatial POIs

To assess computational scalability beyond the 22 curated Padang POIs, a systematic database stress test was conducted in MySQL 8.0. Synthetic destination corpora scaling from \(N = 22\) to \(N = 10,000\) points within the Padang geographic bounding box \([-1.15, -0.80]^\circ\text{ lat}, [100.25, 100.50]^\circ\text{ lng}\) were evaluated over 50 iterations per tier executing full parameterized `ST_Distance_Sphere` bounding queries (`radius <= 20.0 km`, ordered by distance, limit 10). Table 9 outlines the empirical latency progression:

Table 9. Database Scalability and Execution Latency Across Massive POI Corpora

POI Corpus Size (\(N\))	Scale Context	Mean Latency (ms)	Median Latency (ms)	95th Percentile (p95)	Min Latency (ms)	Max Latency (ms)
22	Curated Padang City Baseline	**0.57 ms**	0.48 ms	0.58 ms	0.46 ms	4.14 ms
100	Extended Municipal Attractions	**0.54 ms**	0.52 ms	0.63 ms	0.49 ms	0.70 ms
500	Provincial Tourism Scope	**0.75 ms**	0.74 ms	0.85 ms	0.71 ms	1.12 ms
1,000	Regional Tourism Precinct	**1.02 ms**	1.00 ms	1.17 ms	0.97 ms	1.20 ms

POI Corpus Size (\(N\))	Scale Context	Mean Latency (ms)	Median Latency (ms)	95th Percentile (p95)	Min Latency (ms)	Max Latency (ms)
5,000	Intermediate Synthetic POI Scale (5,000 POIs)	**3.48 ms**	3.13 ms	4.90 ms	3.03 ms	10.84 ms
10,000	Massive Synthetic POI Scale (10,000 POIs)	**6.11 ms**	5.71 ms	10.14 ms	5.51 ms	11.16 ms

As demonstrated in Table 9, **latency increases moderately across the synthetic dataset range** (\(N = 22\) to \(N = 10,000\) POIs, from an average of 0.57 ms to 6.11 ms; 95th percentile of 10.14 ms). Relative to the cumulative cloud LLM inference time across two calls (~1,338 ms), relational spatial computation consumes less than 1.5% of total response duration even at 10,000 destinations. These empirical findings suggest **technical feasibility beyond the current 22-POI dataset**. However, this study intentionally avoids premature claims of turnkey metropolitan-wide readiness, as real-world metropolitan deployment necessitates large-scale multi-user concurrency testing, advanced spatial indexing profiled through explicit execution plans (`EXPLAIN ANALYZE`), and real-time transit integration that lie beyond this single-user synthetic benchmark.

4.8 Secondary Evaluation: Human-Centered Usability and Conversational Interaction (SUS & Cognitive Task Efficiency)

Directly responding to the future research agenda articulated by Afnarius et al. (2026) [2], an empirical user usability evaluation was conducted to assess cognitive load reduction and user satisfaction.

1. Participants and Demographic Profile:

The evaluation enlisted \(N = 30\) independent participants (16 male or 53.33% and 14 female or 46.67%; age range 20–38 years, mean \(24.6 \pm 4.2\) years). Participants represented two distinct target sub-populations: 18 university students in Padang City (tech-savvy digital natives) and 12 independent travelers visiting Padang from outside regions (Riau, Jambi, North Sumatra, and Jakarta).

2. Recruitment and Inclusion Criteria:

Participants were recruited through open announcements at university campuses and regional tourist information centers. Inclusion criteria mandated: (a) active smartphone usage \(\ge 3\) hours daily; (b) regular use of digital mapping applications (Google Maps, OpenStreetMap, Apple Maps) at least once in the preceding month; (c) zero prior exposure to the system prototype or database catalog (*unbiased baseline*); and (d) full fluency in Indonesian.

3. Experimental Design and Counterbalancing (Within-Subjects Latin Square):

A repeated-measures *within-subjects* experimental design was implemented, wherein all 30 participants tested both interfaces: Interface A (conventional WIMP Web GIS featuring dropdown menus, distance sliders, and static modal popups as in *DTExplorer*) and Interface B (Conversational Web GIS driven by the proposed Structured Semantic Control Layer).

To mitigate order effects, learning transfer bias, and cognitive fatigue, a balanced two-group *Latin Square Counterbalancing* protocol was enforced: - **Group 1 (\(N = 15\) participants):** Evaluated Interface A (WIMP) first \(\to\) 15-minute rest interval \(\to\) evaluated Interface B (Chat AI). - **Group 2 (\(N = 15\) participants):** Evaluated Interface B (Chat AI) first \(\to\) 15-minute rest interval \(\to\) evaluated Interface A (WIMP).

4. Standardized Experimental Tasks:

Participants were instructed to execute three standardized travel planning tasks: * **Task 1 (Categorical Discovery):** Discover all beach destinations in Padang City. * **Task 2 (Spatial Proximity Query):** Find Minangkabau culinary destinations located within 5 km from current coordinates. * **Task 3 (Multi-Constraint Complex Planning):** Identify nature destinations with free admission (Rp0) or tickets \(\le \text{Rp15,000}\) that are actively open right now within 10 km.

5. Task Completion Time (TCT) Measurement:

Task Completion Time (TCT) was measured objectively in seconds via screen recording paired with a calibrated digital stopwatch. Timing commenced immediately when the participant finished reading the task prompt and terminated when destination coordinates and navigation routes were completely rendered on the Leaflet.js map. The failed task protocol established a maximum ceiling time-out threshold of **180 seconds (3 minutes)**, whereby any participant unable to retrieve destinations within the threshold would be marked as an unsuccessful task with a recorded time of 180 seconds. In the empirical

trials, all 30 participants successfully completed all three tasks across both interface modalities well within the threshold, yielding a **100% task completion rate**. Table 10 presents the comparative Task Completion Time (TCT) and error rates:

Table 10. Cognitive Task Completion Time (TCT) Comparison: Conventional WIMP vs. Proposed Conversational Web GIS

Benchmark Task	Conventional WIMP Mean TCT (s)	Proposed Conversational Mean TCT (s)	Time Efficiency Gain (%)	WIMP Error Rate (%)	Proposed Error Rate (%)
Task 1: Categorical Filter	18.42 ± 3.15 s	**4.82 ± 0.94 s**	**+73.83%**	3.33%	**0.00%**
Task 2: Proximity Query	34.18 ± 5.60 s	**6.21 ± 1.12 s**	**+81.83%**	10.00%	**0.00%**
Task 3: Multi-Constraint Query	58.74 ± 8.45 s	**8.54 ± 1.35 s**	**+85.46%**	23.33%	**0.00%**
Overall Average	**37.11 ± 5.73 s**	**6.52 ± 1.14 s**	**+82.43%**	**12.22%**	**0.00%**

As outlined in Table 10, the proposed conversational interface reduced average task completion time from **37.11 seconds to 6.52 seconds**, achieving an **82.43% reduction in cognitive interaction duration**. On complex multi-constraint scenarios (Task 3), where users previously suffered a 23.33% error rate from misinterpreting static opening hour tables and adjusting sliders iteratively, the conversational system achieved a **0.00% error rate**.

6. Standardized System Usability Scale (SUS) Scoring:

Following completion of all tasks, participants administered Brooke's standardized 10-item **System Usability Scale (SUS)** [15] questionnaire using a 5-point Likert scale. Composite SUS scores were computed using the standard scoring formula: \[\text{Odd Item Contribution } (X_i) = R_i - 1 \quad (\text{for } i \in \{1, 3, 5, 7, 9\})\] \[\text{Even Item Contribution } (Y_j) = 5 - R_j \quad (\text{for } j \in \{2, 4, 6, 8, 10\})\] \[\text{SUS Composite Score} = 2.5 \times \left(\sum_{i} X_i + \sum_{j} Y_j \right)\] The proposed system attained an aggregate mean SUS score of **84.25 ± 6.80** (median = 85.00; score range 72.50 to 97.50). According to the empirical adjective benchmarks established by Bangor, Kortum, and Miller (2008, 2009), a score of 84.25 resides at a percentile rank \(> 96\%\), Grade "A", and corresponds to an "*Excellent*" rating.

7. Inferential Statistical Significance Testing:

- A normality test conducted on TCT difference values using the *Shapiro-Wilk test* yielded $W = 0.968$ ($p = 0.482 > 0.05$), confirming normal distribution of differences. - A parametric *paired-samples t-test* confirmed that the reduction in task completion time between WIMP (37.11 ± 5.73 s) and Conversational AI (6.52 ± 1.14 s) is statistically significant: $t(29) = 28.42$, $p < 0.001$. The standardized effect size measured by *Cohen's d* reached **5.19** (categorized as a huge effect size). - A non-parametric *Wilcoxon signed-rank test* corroborated this finding with $Z = -4.78$ ($p < 0.001$), establishing the uniform cognitive superiority of the conversational interface across all participants.

8. Research Ethics Clearance:

The experimental usability evaluation protocol was reviewed and granted formal ethics approval by the Research Ethics Committee of LPPM Universitas Andalas (Ethics Approval Letter No. 042/UN16.09/ETIK/2026). All participants provided written informed consent, participated voluntarily without coercion, were assured full data anonymity, and retained the unconditional right to withdraw at any stage without prejudice.

9. Open Raw Data Availability:

In alignment with open science and empirical audit standards, raw item-by-item SUS Likert scores (10 items), precise TCT duration measurements per task, participant demographics, and counterbalancing sequence logs are publicly accessible in `jurnal/LAMPIRAN_RAW_DATA_SUS_TCT.md` and in structured tabular format at `jurnal/usability_sus_tct_30_respondents.csv`.

4.9 Spatial Transferability, OSRM Service Provenance, and System Limitations

To address critical peer review criteria regarding geographic transferability and empirical reproducibility, architectural transferability and external service dependencies are detailed as follows:

1. Geographic Portability with Limited Domain-Specific Reconfiguration:

Rather than asserting absolute zero-code portability, this study realistically defines that the **architecture is designed to support geographic portability with limited domain-specific reconfiguration**. Core components that are intrinsically domain-agnostic and reusable without source code modifications comprise: the Canonical Spatial Intent Representation (CSIR) schema, the 6-dimensional *SirValidator* rule set, the *Deterministic*

Spatial Query Compiler, the mathematical spatial distance formula (`ST_Distance_Sphere`), and the *Claim-Level Grounding Validator*. However, transferring the framework to a new urban or regional municipality necessitates five domain-specific reconfigurations: - Updating bounding box geographic coordinates and registering relational POI catalog entries in MySQL. - Adapting local tourism category ontologies (e.g., introducing categories such as royal heritage or artisanal craft districts). - Adjusting local administrative hierarchy classifications (sub-districts, districts, or rural villages). - Updating out-of-scope negative vocabulary lists tailored to the target city's physical geography. - Configuring local operating hours semantics and OSRM vehicular routing profiles.

2. Positioning of OSRM Service and External Computation Boundaries:

The architecture enforces a strict functional bifurcation between factual spatial computation and trajectory visualization: - **MySQL 8.0 Spatial Engine (`ST_Distance_Sphere`):** Executes 100% of factual spatial computations, mathematical radius bounding, and deterministic candidate proximity ordering in-database (1.21 ms latency). This constitutes the core spatial algorithmic contribution. - **Open Source Routing Machine (OSRM):** Positioned strictly as an auxiliary presentation service for rendering turn-by-turn road network polyline vectors and trip duration estimates on the Leaflet.js canvas. - **Data Provenance and External Dependencies:** Road network geometries originate from OpenStreetMap (OSM) released under the Open Database License (ODbL). Reliance on public OSRM endpoints (`router.project-osrm.org`) introduces external dependencies relevant to benchmark replication: (a) dynamic OSM community edits; (b) public server rate limiting; and (c) external network latency (~88.40 ms, range 45.20–142.50 ms) residing outside municipal server control. - **Architectural Fault Tolerance:** The system encapsulates external service failure. Should the external OSRM endpoint experience downtime or network throttling, spatial discovery, candidate retrieval, and grounded NLG synthesis remain 100% operational; the Leaflet.js client interface displays a non-blocking route failure toast notification while omitting the road polyline, leaving POI markers and all relational database facts intact.

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5. CONCLUSION AND FUTURE WORK

This study designed, implemented, and empirically evaluated a **Reliability-Controlled Conversational Spatial Information System** governed by an architectural framework of **Dual Control Boundaries**, validated across 22 curated tourism destinations in Padang City. Expanding upon the exploratory spatial interaction lineage of *DTExplorer* (Afnarius et al., 2026), the architecture establishes two explicit control boundaries that decouple non-deterministic, probabilistic natural language interpretation from deterministic spatial computation, harnessing generative AI flexibility while guaranteeing mathematical and factual data integrity.

Empirical evaluation across 40 standardized conversational scenarios, 34 automated unit tests (112 assertions, 100% pass), a 10,000-POI scalability stress test, and an empirical usability trial ($N = 30$) yielded the following findings: 1. **Efficacy of the Semantic Control Boundary:** The isolation of LLM parsing into a typed **Canonical Spatial Intent Representation (CSIR)** governed by the *Spatial Operator Ontology*, verified by the six-dimensional **SIR Validator** enforcing the *No Intent Alteration* principle, achieved an Exact CSIR semantic extraction accuracy of **92.50%** (37/40) for pure user intent, a **90.00%** (36/40) post-policy execution envelope alignment, an average field-level accuracy of **99.25% (398/401)**, a **95.00%** category classification accuracy (38/40), a **100.00%** pure spatial predicate correctness (36/36, alongside 100.00% / 40/40 overall scenario compliance), a **100.00%** validation decision accuracy (40/40), and a **100.00% (2/2)** Honest Rejection Rate on tested out-of-scope requests, with zero unverified or silently mutated parameters reaching database compilation. 2. **Efficacy of the Evidence Control Boundary and Relational Spatial Execution:** The relational database (MySQL 8.0 `ST_Distance_Sphere`) established the sole source of spatial truth, executing spherical distance queries in an average of **1.21 ms** (scaling to 6.11 ms under 10,000 synthetic POIs). Enforcing the *Strict Grounding Contract* paired with post-generation algorithmic auditing by the **Claim-Level Grounding Validator** across 4 sub-dimensions ($\mathcal{C}_{\text{entity}}, \mathcal{C}_{\text{price}}, \mathcal{C}_{\text{spatial}}, \mathcal{C}_{\text{temporal}}$) achieved **100.00% Claim-Level Grounding Fidelity (304/304 claims)**, an **Entity Grounding Fidelity of 100.00% (147/147 POI mentions)**, a **100.00% Scenario Grounding Pass Rate (40/40 scenarios)**, and **0 observed fabricated POIs (0.00% Entity Fabrication Rate)** under the evaluated benchmark conditions, fortified by deterministic fail-closed fallback mechanisms. 3. **Technical Latency and Usability Excellence:** The system recorded an

average server-side conversational pipeline latency of **1,252.17 ms** (median: 1,232.39 ms, p95: 1,581.37 ms) and an average end-to-end user-perceived response latency of **1,340.57 ms** (median: 1,316.89 ms, p95: 1,709.97 ms) incorporating independent client-side OSRM road routing (88.40 ms). Empirically, the conversational paradigm demonstrated an **82.43% reduction in cognitive task completion time** over conventional WIMP interfaces (\(t(29) = 28.42, p < 0.001\), Cohen's \(d = 5.19\)), and achieved an "Excellent" **System Usability Scale (SUS) score of 84.25 ± 6.80**. Architectural robustness was confirmed by a 100% pass rate across 34 automated unit tests (112 assertions).

Future work will focus on extending the control layer to multi-turn dialogue state tracking across dynamic re-routing, expanding query capabilities to multilingual international tourist dialogues, and deploying the framework across provincial multi-destination corridors.

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APPENDIX (SUPPLEMENTARY MATERIAL)

APPENDIX A: Formal System Prompt Specification for Spatial Intent Representation (SIR) Extraction

The following complete five-layered declarative system prompt is injected into Layer 2 (*Semantic Interpretation Layer*) to strictly bound language model cognitive operations (`temperature: 0.0`, `response_format: {"type": "json_object"}`):

```
ROLE:
You are a spatial intent parser for the tourism Web GIS of Padang City, Indonesia.

TASK:
Transform the user's natural-language query into exactly one structured Spatial Intent Representation (SIR) in pure JSON.

OUTPUT CONTRACT:
1. Return valid JSON only. No markdown formatting, no code fences, no explanatory text.
2. Return strictly the defined schema fields.
3. Do not generate SQL queries, database clauses, or table names.
4. Do not answer the user's question, do not converse, and do not provide recommendations.
5. Do not invent or recommend tourism objects.

SEMANTIC RULES:
6. category: Must use ONLY one of the supported categories: "Pantai" | "Pulau" | "Alam" | "Museum" | "Sejarah" | "Kuliner" | null.
7. spatial_operator: Must use ONLY one of: "nearest" | "within_radius" | "within_admin_area" | "none".
8. target_name: Represents a specific POI explicitly named by the user (e.g., "Pantai Air Manis"). Otherwise null.
9. keyword: Represents descriptive search terms or features (e.g., "pasir putih", "snorkeling"). Never convert a keyword into a target_name.
10. Do not resolve a user-mentioned name to a database POI ID.
11. Do not invent coordinates, distances, prices, opening hours, or administrative areas.

SPATIAL & PRESERVATION RULES:
12. Preserve spatial constraints exactly as expressed by the user.
13. If a spatial operator requires a radius but the user did not specify one, set radius: null. NEVER infer, guess, or default a radius value.
14. Do not convert or silently modify a user's spatial constraint (e.g., do not clamp or modify negative numbers).

PRICE & BUDGET RULES:
15. is_free: Set to true ONLY if the user explicitly requests free admission ("gratis", "free").
16. max_price: Represents the explicit upper price ceiling specified by the user (integer in IDR). If no price is mentioned, set to null. Do not infer a price.

TEMPORAL RULES:
17. open_now: Set to true ONLY when the user explicitly requests currently open/operating places ("buka sekarang", "sedang buka").
18. open_24h: Set to true ONLY when the user explicitly requests 24-hour operation ("24 jam").

REFERENCE RULES:
19. GPS coordinates are supplied strictly by the application context; NEVER infer or invent latitude or longitude coordinates.
20. reference_type: Use "gps" when user refers to current location ("dekat saya", "from here"), "city_center" for city center, "poi" when referencing
```

```
another POI, or "unknown" if unspecified.

SORTING RULES:
21. sort: Use ONLY "termurah" (lowest price) | "termahal" (highest price) |
"terdekat" (nearest distance) | "terbaik" (highest public review rating) |
null.

SCOPE RULES:
22. Set is_out_of_scope: true when the request requires an activity,
entity, or geographic location outside the Padang tourism domain (e.g.,
ski, snow, casino, destinations in other cities like Borobudur/Bali).
23. Do not treat missing database information as out of scope.

FINAL RULE:
24. When information is missing or ambiguous, preserve uncertainty in the
SIR (using null) rather than guessing.

SCHEMA:
{
  "intent": "spatial_recommendation" | "entity_lookup" | "general_inquiry",
  "entity": "tourism_object",
    "category": "Pantai" | "Pulau" | "Alam" | "Museum" | "Sejarah" |
"Kuliner" | null,
  "target_name": string or null,
  "keyword": string or null,
   "spatial_operator": "nearest" | "within_radius" | "within_admin_area" |
"none",
  "reference_type": "gps" | "city_center" | "poi" | "unknown",
  "reference_entity": string or null,
  "radius": float or null,
  "distance_unit": "km",
  "admin_area": string or null,
  "is_free": boolean,
  "max_price": integer or null,
  "open_now": boolean,
  "open_24h": boolean,
  "sort": "termurah" | "termahal" | "terdekat" | "terbaik" | null,
  "is_out_of_scope": boolean
}
```

APPENDIX B: Formal System Prompt Specification for Grounded Natural Language Generation (Grounded NLG Contract)

The following complete system prompt is injected into Layer 5 (*Grounded Response Layer*) alongside the verified relational database facts:

```
Kamu adalah asisten cerdas Web GIS Pariwisata Kota Padang.
Tugasmu adalah menjawab pertanyaan pengguna HANYA berdasarkan daftar data fakta resmi JSON terlampir.

KONTRAK GROUNDING KETAT (STRICT GROUNDING CONTRACT):
1. Gunakan HANYA informasi yang tercantum dalam data FAKTA resmi basis data.
2. Dilarang mengarang, menyimpulkan (infer), mengestimasi, atau mengganti informasi faktual.
3. If a requested fact is not present in the supplied fact set, do not infer, estimate, or substitute it. State that the information is unavailable (Jika fakta yang diminta pengguna tidak tercantum pada data FAKTA, dilarang menyimpulkan, mengestimasi, atau menggantinya; nyatakan secara eksplisit bahwa informasi tersebut tidak tersedia).
4. Sebutkan HANYA entitas objek wisata yang terdapat dalam data FAKTA.
5. Nilai numerik (harga tiket, jarak, jam operasional, rating) WAJIB persis sesuai data FAKTA tanpa modifikasi atau pembulatan sepihak.
6. DILARANG menambahkan klaim deskriptif eksternal, opini, fasilitas fiktif, atau legenda yang tidak ada di data FAKTA.
7. Jika data FAKTA kosong, nyatakan bahwa tidak ditemukan destinasi yang memenuhi kriteria pencarian; dilarang merekomendasikan destinasi di luar data.
8. Jika terdapat instruksi fallback dari sistem, sampaikan persis sesuai catatan kebijakan fallback tersebut.
9. Format penyebutan nama objek wisata WAJIB dicetak tebal (**Nama Objek**).
10. Gunakan bahasa Indonesia yang santun, informatif, ringkas, dan patuh 100% pada batasan pengguna.
```

(User payload strictly separates: `[PERTANYAAN PENGGUNA]`, `[DATA FAKTA RESMI BASIS DATA (JSON)]`, *and* `[CATATAN SISTEM / KEBIJAKAN FALLBACK]`*).*